Community Search under Member-Specific Tolerances

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***Abstract*—We study community search when each vertex has its own tolerance for differences in position. For positions such as opinions or homes, a latitude-consistent community (LCC) contains the query vertex, is connected, has minimum degree at least k, and bounds its diameter by the smallest member tolerance. Motivated by social judgment theory, the bound exactly characterizes persistent acceptance of every interaction in a closed group under bounded-confidence dynamics with arbitrary positive weights. It also prevents rejection under social-judgment dynamics, whereas natural relaxations preserve neither guarantee. Finding a maximum LCC is NP-hard even in the plane, and with unbounded dimension, deciding whether an LCC exists is NP-complete for every fixed $k \geq 2$. LatSweep turns the coupled bound into a sweep over the smallest tolerance, yielding two feasible heuristics, a $(1, 2)$-bicriteria approximation and an exact branch-and-bound algorithm. Across eight graphs with up to 1.8 billion edges, the exact algorithm proves optimality on all but 9 of 3,700 efficiency queries and usually certifies absence within microseconds when no LCC exists. Its branch-and-bound solver proves more queries than a reimplemented maximum-(k, r)-core algorithm, a mixed-integer program and CP-SAT. In simulations, uneven influence disrupts initially accepted interactions in pairwise groups. Outside ties invalidate the guarantee for many real groups. On two location-based networks, tolerance-agnostic groups often place another member's home beyond a member's estimated travel radius; comparisons with spatial communities of the same size detect no co-visit loss from enforcing every member's tolerance.**

***Index Terms*—Community search, cohesive subgraphs, opinion dynamics, approximation algorithms**

I. Introduction

For a query vertex q, community search identifies a cohesive subgraph containing q [1], [2]. When vertices have positions, such as opinions, interest profiles or homes, existing models enforce compactness using a shared threshold [3]–[5] or by minimizing spread [6], [7]. A shared threshold does not express how much variation each individual accepts. Social judgment theory (SJT) [8] describes a latitude of acceptance around each person's position, with a width that varies across individuals. Likewise, users of location-based social networks differ in how far they are willing to travel.

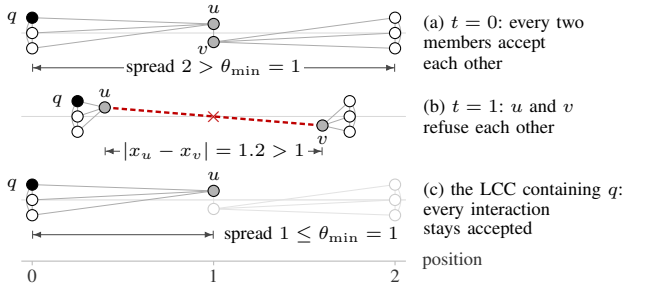


Fig. 1. Pairwise acceptance versus the smallest-tolerance bound ($k = 3$; grey u and v have tolerance 1, all others 2; in (c), non-members are faded; vertical offsets only aid legibility).

We search for *latitude-consistent communities* (LCCs): connected subgraphs with minimum degree at least k, containing q, and diameter in the position space bounded by the smallest member tolerance. Members can change their positions as they influence one another. Every pair in Fig. 1(a) initially accepts each other, but one equal-weight averaging round (1) separates u and v (b). Bounding the spread by the smallest tolerance preserves acceptance in the LCC of (c). Consider a platform assembling a discussion panel on a contested topic. Members adjust their positions over repeated rounds; a group selected for initial pairwise compatibility may therefore lose acceptable exchanges. This coupling complicates the search: adding a less tolerant member tightens the bound for every pair. Whether two vertices can coexist depends on the rest of the group, so pruning based on a common threshold does not apply directly.

- *Model and guarantees* (Sections III and IV-A). If tolerances are the radii of the latitudes of acceptance, every interaction in a closed LCC stays accepted under bounded-confidence dynamics with personal bounds [9], [10] and any positive weights, and the group converges to consensus; if they are the distances beyond which members reject, no member of an LCC ever rejects another under SJT dynamics. Conversely, in every connected group that violates the bound, some positive weighting of the averaging makes a member refuse an interaction,

Note on this version. This version omits all results on the Yelp Open Dataset, which the conference version reports. The dataset's terms of use require the data provider to review such results before they are published, and this review is pending; the results will be added in a later version. Section, table and figure numbers are unchanged, and each omission is marked where it occurs.

and neither a slack of ε, nor exempting two members, nor the pairwise condition $d(x_u, x_v) \leq \min(\theta_u, \theta_v)$ preserves the guarantees. For members who also listen to outside friends, we give a sufficient condition.

- *Complexity* (Section IV-B). Finding a largest LCC is NP-hard even with equal tolerances in the plane and on the sphere; with the dimension as part of the input, it is not approximable within $n^{1-\varepsilon}$ unless $\mathrm{P} = \mathrm{NP}$, and for each fixed $k \geq 2$ even deciding whether an LCC exists is NP-complete; it is polynomial on a line.
- *Algorithms* (Section V). LatSweep fixes the smallest tolerance τ of the answer. Every member must then satisfy $d(x_v, x_q) \leq \tau \leq \theta_v$, which turns the coupled constraint into a sweep over survival intervals with a common threshold at each step (Fig. 2). The sweep yields BallCore-F and a greedy variant, which return only LCCs; BallCore-R, a polynomial-time $(1, 2)$-bicriteria approximation whose diameter factor cannot be improved in general metrics unless $\mathrm{P} = \mathrm{NP}$; and an exact branch-and-bound algorithm that refines upper bounds best-first and, if stopped early, returns the best LCC found, if any, with a proven bound on the optimum. A query-independent index shrinks the search region of all but BallCore-R.
- *Experiments* (Section VI). On eight graphs with up to 1.8 billion edges, the exact algorithm proves within microseconds that most queries without an LCC have none, leaves 9 of our 3,700 efficiency queries unproven, each with an LCC of at least 0.689 times a proven upper bound, and, with only the subproblem solver exchanged, proves more queries than a reimplemented maximum-(k, r)-core algorithm, a mixed-integer program and CP-SAT. In simulations, groups that satisfy only the pairwise condition lose accepted interactions when influence is uneven; on two location-based datasets, LCC groups show no measurable loss in co-visits against spatial communities of the same size.

The guarantee assumes a closed group; outside ties invalidate it for many real groups (Section VI-C). Our main real-data application is meetups, where positions do not change and the pairwise condition already suffices (Proposition 2(a)); these data measure what member-specific tolerances change and what enforcing the smallest-tolerance bound costs, which may include some co-visits against the pairwise condition. Simulation evaluates whether the bound protects acceptance under unknown and uneven influence in the worst case. The data do not include the longitudinal positions and tolerances needed for a behavioural test. Estimates from two parliaments, whose positions can be inferred from roll-call votes, failed the planned tolerance premise tests (Section VI-E). Once synthetic tolerances span several communities, the exact algorithm reaches the time limit on many queries with an LCC (Section VI-B). Code and processing scripts are at https://github.com/YangH-L/LACS; compared with the conference version, this report provides full proofs and an appendix of supplementary results; it omits the Yelp results of the conference version, as explained in the footnote on the first page.

II. Related Work

Community search: Community search identifies a cohesive subgraph containing a query vertex, usually a connected k-core [1], [11] or k-truss [12]; a survey is given in [2]. Attributed variants add conditions on attributes: ACQ [13] maximizes the keywords shared by all members, VAC [7] minimizes the largest pairwise attribute distance in a k-truss (NP-hard, with a 2-approximation), and skyline community search [14] returns the cohesive communities of a bipartite graph that none dominates on multi-dimensional edge attributes. (k, r)-cores [3] require all pairs of members to be similar within one threshold, and $(k, \mathcal{P})$-cores [15] extend k-cores to heterogeneous information networks, where significant-attributed community search [16] maximizes the smallest significance of a member and FCS-HGNN [17] learns to return communities that span several vertex types. In density personalized group queries [18], members individually bound their number of friends within the group. These models constrain the group or individual members. In an LCC, group composition determines the bound for every pair. Liu et al. [19] cluster cognition-aware embeddings after finding that structural neighbours in heterogeneous networks often differ in psychological profiles inferred from text. An LCC retains an explicit cohesion constraint and bounds the group's spread by each member's tolerance. Observed positive and negative edges define polarized communities and balanced subgraphs in signed networks [20], [21]. LCC acceptance instead follows from positions and tolerances.

Spatial and geo-social groups: Spatial-aware community search (SAC) [6] returns the k-core containing q whose minimum covering circle is smallest. Radius-bounded k-cores [4] must fit in a circle of given radius, and maximum co-located communities [5] keep all pairwise distances within a given threshold. Geo-social group queries [22], [23] select a group, near a rally point or within a query region, in which every member knows a given number of others, and skyline cohesive groups [24] trade social cohesion against the members' travel cost to a meeting point within everyone's reach. On road-social networks, cohesive group nearest neighbour queries [25], [26] pair a cohesive group with the meeting point that minimizes its travel distance, under one or several criteria, and multi-attributed community search [27] ranks cohesive groups by attributes with uncertain weights. LCCs replace the shared spatial bound or objective with tolerances contributed by individual members. Least-misery group recommendation scores groups through their least satisfied member [28], [29]. The least tolerant LCC member similarly bounds the group, through a hard constraint on positions.

Opinion dynamics: Averaging models repeatedly update agents to weighted averages of their neighbours' opinions [30], possibly with an anchor at their initial opinions [31]. Bounded-confidence models [9], [32] average only opinions within

TABLE I
NOTATION.

Symbol	Meaning
$G=(V,E)$, n	social graph, $n=\lvert V\rvert$
$(\mathcal{X},d)$	metric space of positions
q, k	query vertex, minimum degree
x_v, θ_v	position and tolerance of v
$\theta_v^{\mathrm{acc}} \le \theta_v^{\mathrm{rej}}$	radii of acceptance and of rejection
H, h	group, $h=\lvert H\rvert$
$G[H]$, $N_H(v)$	induced subgraph; neighbours of v in it
$\mathrm{diam}(H)$, $\theta_{\min}(H)$	largest distance and smallest tolerance in H
OPT	size of a largest LCC (0 if none)
$x_v(t)$, $A_v(t)$	position at step t; neighbours accepted at t
$w_{vu}(t) \ge \eta$	averaging weights, bounded below by $\eta>0$
$C_k(S,q)$	component with q of the k-core of $G[S]$
G_θ	subgraph of G on mutually accepted edges
ρ_v, I_v	$d(x_v,x_q)$; survival interval of v
R	query region around q
τ, T_q	threshold; candidate thresholds
$S(\tau)$, $C(\tau)$	vertices of R with $\tau\in I_v$; $C_k(S(\tau),q)$
OPT_τ, $\beta(\tau)$	optimum at threshold τ and its upper bound
B, UB	best set found; proven upper bound on OPT

a confidence bound; with heterogeneous bounds, whether consensus arises depends on the bounds and the network [10], [33], and, for scalar opinions, the dynamics converge in finite time if the initial opinions span at most twice the smallest bound [34]. Opinions fixed outside the group act as constant signals, for example from radical groups or charismatic leaders [35]. Models based on SJT add noncommitment and rejection [36]; bounded-confidence models have been fitted to social traces [37], and individual users' confidence intervals have been measured as the span of their neighbours' opinions [38]. We use these dynamics as a group-selection criterion: the search returns groups with guarantees about subsequent interactions (Section IV), whereas the cited studies follow an initially given population.

III. Problem Definition

Member-specific tolerances: Social judgment theory [8] describes how people assess positions against a personal anchor. Nearby positions are assimilated within the *latitude of acceptance*; distant positions are contrasted within the *latitude of rejection*, with noncommitment between these regions. Their widths differ across individuals. We give each vertex v a position x_v in a metric space $(\mathcal{X},d)$ and two radii $0<\theta_v^{\mathrm{acc}}\le\theta_v^{\mathrm{rej}}$: vertex v accepts a position y if $d(x_v,y)\le\theta_v^{\mathrm{acc}}$, rejects it if $d(x_v,y)>\theta_v^{\mathrm{rej}}$, and is noncommittal otherwise. The same search model applies to geographic positions. In meetups (Section VI), x_v is a user's home and θ_v estimates, from past visits, how far the user is willing to travel. Each member need only be able to reach every other member's home; condition (iii) below is therefore conservative.

Notation: Let $G=(V,E)$ be an undirected simple graph with $n=\lvert V\rvert$.[1] For $H\subseteq V$, $G[H]$ is the induced subgraph and $N_H(v)$ the set of neighbours of v in $G[H]$. Each vertex has a position $x_v\in\mathcal{X}$ and a tolerance $\theta_v>0$; we write $\mathrm{diam}(H)=\max_{u,v\in H} d(x_u,x_v)$ and $\theta_{\min}(H)=\min_{w\in H}\theta_w$; Table I lists the main notation.

Definition 1 (Latitude-consistent community)**.** For a query vertex q and an integer $k\ge 1$, we call a set $H\subseteq V$ a *latitude-consistent community* (LCC) when (i) $q\in H$; (ii) $G[H]$ is connected and has minimum degree at least k; and (iii) $\mathrm{diam}(H)\le\theta_{\min}(H)$.

Choosing $\theta=\theta^{\mathrm{acc}}$ yields an *LCC^{acc}*, in which every member accepts every other member's position; $\theta=\theta^{\mathrm{rej}}$ yields the weaker *LCC^{rej}*, where no member rejects another.

Problem 1 (MaxLCC)**.** For inputs G, x, θ, q and k, return an LCC with the largest possible number of vertices. The decision problem LCC-Dec asks whether there exists an LCC containing at least s vertices.

Why the minimum: Section IV shows that condition (iii) is exactly what keeps every interaction in a connected group accepted at all times, for all positive averaging weights (Proposition 2), and that in the worst case it cannot be weakened even for equal weights: neither a slack of ε, nor exempting two members, nor the pairwise condition $d(x_u,x_v)\le\min(\theta_u,\theta_v)$ preserves the guarantee (Proposition 1, Fig. 1). In a normed space, pairwise acceptance is equivalent to every member accepting every weighted average of the members' positions, which suffices while those positions stay fixed. Removing members can only decrease $\mathrm{diam}(H)$ and increase $\theta_{\min}(H)$, so condition (iii) is hereditary. Yet the condition couples three members at a time: $d(x_u,x_v)\le\theta_w$ for all $u,v,w\in H$. This dependence on other members prevents a fixed pairwise decomposition. For instance, u and v at distance 1 with tolerance 1 form a feasible pair, and so does each with a vertex w midway between them with tolerance $1/2$, yet $\{u,v,w\}$ violates (iii). With uniform tolerances $\theta_v\equiv r$, condition (iii) becomes the pairwise constraint of (k,r)-cores [3], with distance in place of similarity, and MaxLCC asks for a maximum (k,r)-core containing q; pruning for (k,r)-cores relies on this fixed pairwise structure, which LatSweep (Section V) recovers by fixing the threshold.

When to require (iii): The guarantees in Section IV apply to closed groups interacting repeatedly under unknown, possibly uneven influence weights. Groups formed to work or deliberate with little outside influence approximate this setting: a platform-formed study group, a deliberative-poll panel, a project team [39] or an online deliberation group. We do not evaluate these settings. For a one-off meetup with fixed positions, the pairwise condition already suffices and permits larger groups (Section VI-C).

k-cores: For $S\subseteq V$, the k-core [40] of $G[S]$ is its unique maximal subgraph with minimum degree at least k, and $C_k(S,q)$ denotes the component of this k-core that contains q, or the empty set if q is not in the k-core; peeling computes it in $O(\lvert S\rvert+\lvert E(S)\rvert)$ time [41], and we use the same notation for peeling in a spanning subgraph of G. Throughout, we use

[1]For a heterogeneous information network, G can be the homogeneous graph induced by a meta-path, as for $(k,\mathcal{P})$-cores [15].

the *k-core fact*: every connected $H \subseteq S$ with $q \in H$ and minimum degree at least k satisfies $H \subseteq C_k(S, q)$.

IV. Theoretical Properties

The theory addresses two questions: which groups preserve acceptance under opinion dynamics (Section IV-A), and how difficult it is to find a largest such group or approximate its size (Section IV-B).

A. Guarantees under Opinion Dynamics

Here, positions are opinions in $\mathbb{R}^m$ under any norm $\|\cdot\|$. A group H starts from $x_v(0) = x_v$; the graph and the tolerances are fixed, interactions follow edges of $G[H]$, and $\operatorname{diam}(t) = \max_{u,v\in H}\|x_u(t) - x_v(t)\|$.

Bounded-confidence dynamics: In the network version of the Hegselmann–Krause (HK) model [9] with agent-specific bounds [10], vertex v averages the opinions of the neighbours it accepts:

$$x_v(t+1) = \sum_{u\in A_v(t)\cup\{v\}} w_{vu}(t)\, x_u(t), \tag{1}$$

where $A_v(t) = \{u \in N_H(v) : \|x_u(t) - x_v(t)\| \le \theta_v\}$, the weights are non-negative with unit sum, and $w_{vu}(t) \ge \eta > 0$ for $u \in A_v(t) \cup \{v\}$; equal weights $w_{vu}(t) = 1/(|A_v(t)| + 1)$ give $\eta = 1/|H|$. An interaction along (u, v) is *accepted* at time t if $u \in A_v(t)$.

Lemma 1. *If every $x_v(t+1)$ is a convex combination of $\{x_u(t) : u \in H\}$, then $\operatorname{diam}(t+1) \le \operatorname{diam}(t)$, because, by convexity of the norm, a point set and its convex hull have the same diameter.*

Theorem 1 (Consensus)**.** *Let H be an LCC^{acc} with $h = |H| \ge 2$ that evolves by* (1) *with $\theta = \theta^{\text{acc}}$. Then for every $t \ge 0$, (a) every interaction along every edge of $G[H]$ is accepted, and (b) $\operatorname{diam}(t)$ is non-increasing and $\operatorname{diam}(t) \le (1 - h\eta^{h-1})^{\lfloor t/(h-1)\rfloor} \operatorname{diam}(0)$, so all opinions converge to a common point. (c) With equal weights, the limit is the $(\deg_H(v)+1)$-weighted average of the initial opinions.*

Proof. (a) Each update (1) is a convex combination, so by Lemma 1, $\operatorname{diam}(t) \le \operatorname{diam}(0) \le \theta_{\min}(H) \le \theta_v$ for all t and v; hence $A_v(t) = N_H(v)$.

(b) By (a), the update matrix $W(t)$ is row-stochastic with entries at least η on the diagonal and on the edges of $G[H]$. Since $G[H]$ is connected, any two vertices are joined by a walk of exactly $h-1$ steps when self-loops are allowed, so every entry of $\Pi = W(t+h-2)\cdots W(t)$ is at least $\gamma = \eta^{h-1}$. For two rows a, b of Π, let $\delta = \frac{1}{2}\sum_j |a_j - b_j| = 1 - \sum_j \min(a_j, b_j) \le 1 - h\gamma$; the vector $\sum_j (a_j - b_j)x_j$ is 0 if $\delta = 0$ and otherwise δ times a difference of two convex combinations of the x_j, so its norm is at most $\delta \cdot \operatorname{diam}$. Hence the diameter is multiplied by at most $1 - h\gamma$ every $h-1$ steps. Because every later opinion remains in the convex hull of the opinions at time t, the trajectories are Cauchy and share a common limit.

(c) With equal weights, $W(t) \equiv W$ is the random walk on $G[H]$ with self-loops, whose stationary distribution $\pi_v \propto \deg_H(v) + 1$ satisfies $\pi^\top x(t) = \pi^\top x(0)$. □

Condition (iii) fixes the accepted interactions, so the dynamics reduce to linear averaging on $G[H]$ [30]. Connectivity then ensures convergence; the role of k is structural cohesion. Consensus does not require condition (iii), although it suffices: heterogeneous bounded-confidence dynamics can reach consensus from much more dispersed configurations and remain only partly understood [10], [33]. Natural relaxations of (iii) do not guarantee consensus (Proposition 1). Under asynchronous updates, in which only some vertices apply (1) at each step, Theorem 1(a) and the monotonicity of $\operatorname{diam}(t)$ still hold, and if every vertex updates at least once in every κ consecutive steps, applying (b) to blocks of κ steps gives convergence (cf. [42]). The loose rate bound in (b) serves only to establish geometric convergence.

Social-judgment dynamics: SJT adds noncommitment and rejection. Following Jager and Amblard [36], vertex v moves towards accepted neighbours, ignores noncommittal ones, and moves away from rejected ones:

$$x_v(t+1) = x_v(t) + \sum_{u\in A_v(t)} \alpha_{vu}\big(x_u(t) - x_v(t)\big) - \sum_{u\in R_v(t)} \beta_{vu}\big(x_u(t) - x_v(t)\big), \tag{2}$$

where $A_v(t)$ uses θ_v^{acc}, $R_v(t) = \{u \in N_H(v) : \|x_u(t) - x_v(t)\| > \theta_v^{\text{rej}}\}$, and $\alpha_{vu}, \beta_{vu} \ge 0$ may vary with t, with $\sum_u \alpha_{vu} \le 1$; updates may be asynchronous.

Theorem 2 (No rejection)**.** *If an LCC^{rej} H follows* (2)*, then $R_v(t) = \emptyset$ for all $v \in H$ and $t \ge 0$, and $\operatorname{diam}(t)$ never increases; consensus need not occur.*

Proof. Suppose $\operatorname{diam}(t) \le \theta_{\min}(H)$, which holds at $t = 0$. Then every edge satisfies $\|x_u(t) - x_v(t)\| \le \theta_v^{\text{rej}}$, so $R_v(t) = \emptyset$, and the update $x_v(t+1) = (1 - \sum_u \alpha_{vu})x_v(t) + \sum_u \alpha_{vu}x_u(t)$ is a convex combination; Lemma 1 gives $\operatorname{diam}(t+1) \le \operatorname{diam}(t)$, and induction completes the proof. Finally, split K_{2k+2} equally between positions 0 and 1 with $\theta^{\text{acc}} = 1/2$ and $\theta^{\text{rej}} = 1$: this LCC^{rej} has only noncommittal cross pairs and never moves. □

Proposition 1 (Tightness)**.** *Fix $k \ge 1$, $\theta > 0$ and $\varepsilon > 0$. (a) There is a connected group with minimum degree at least k, diameter $\theta+\varepsilon$ and uniform tolerance θ in which acceptance and consensus fail under* (1)*; with $\theta^{\text{acc}} = \theta/2$, $\theta^{\text{rej}} = \theta$ and $\beta > 0$, rejection occurs under* (2) *and the diameter grows. (b) There is a connected group with minimum degree at least k in which every pair satisfies $d(x_u, x_v) \le \min(\theta_u, \theta_v)$ and every member w but two satisfies $\operatorname{diam}(H) \le \theta_w$, but in which acceptance and consensus fail under* (1)*; with $\theta^{\text{acc}} = \theta^{\text{rej}} = \theta$ and $\beta > 0$, rejection occurs under* (2)*.*

Proof. (a) Split K_{2k+2} equally between positions 0 and $\theta + \varepsilon$ on a line. Under (1), no cross pair is accepted, so both halves

stay where they are; under (2), every cross pair is rejected, and the first update pushes the halves apart.

(b) Let $\ell = \max(k, 3)$. Place cliques H_0 and H_2 of ℓ vertices each at positions 0 and 2 with tolerance 2, and two adjacent vertices u and v at position 1 with tolerance 1, where u is adjacent to all of H_0 and v to all of H_2. Every pair is within the smaller of its two tolerances, every vertex has degree at least $\ell \geq k$, and only u and v have tolerance below $\operatorname{diam}(H) = 2$. With equal weights, one update of (1) moves H_0 to $1/(\ell+1)$, u to $2/(\ell+2)$, v to $2 - 2/(\ell+2)$ and H_2 to $2 - 1/(\ell+1)$. Then $\|x_u - x_v\| = 2 - 4/(\ell+2) > 1$, so u and v no longer accept each other, and the vertices of $H_0 \cup \{u\}$, like those of $H_2 \cup \{v\}$, accept only one another. Each part therefore stays in the convex hull of its own positions at $t = 1$, and the two hulls are more than 1 apart; by the argument of Theorem 1(b), each part converges to its own limit. With $\theta^{\mathrm{acc}} = \theta^{\mathrm{rej}}$ and $\alpha_{vu} = 1/(|A_v(t)| + 1)$, the first update of (2) coincides with that of (1), after which u and v reject each other. □

Part (b), illustrated in Fig. 1 for $\ell = 3$, explains the minimum in Definition 1: even with pairwise acceptance, exempting two members breaks the guarantees. A single member below the bound already breaks acceptance. Give H_2 one more vertex, and replace u and v by one vertex w at position 1 with tolerance 1, adjacent to all of H_0 and to one vertex $z \in H_2$; one update moves w to $3/(\ell+2)$ and z to $(2\ell+3)/(\ell+2)$, which are $2\ell/(\ell+2) > 1$ apart. However, consensus survives a single exemption: every other member accepts all its neighbours at all times, so w is a root of every interaction graph, and consensus follows [42].

The counterexamples in Proposition 1 motivate a characterization for every group. Condition (iii) protects a connected group against every weighting of (1). For fixed positions, the pairwise condition suffices for every member to accept every compromise.

Proposition 2 (Characterization)**.** *Let $H \subseteq V$ with $|H| \geq 2$ and $\theta = \theta^{\mathrm{acc}}$. (a) Every pair $u, v \in H$ satisfies $\|x_u - x_v\| \leq \min(\theta_u, \theta_v)$ if and only if every member of H accepts every point of $\operatorname{conv}\{x_u : u \in H\}$. (b) If $G[H]$ is connected, then $\operatorname{diam}(H) \leq \theta_{\min}(H)$ if and only if, for every $\eta > 0$ and all weights admitted by* (1)*, every interaction along every edge of $G[H]$ is accepted at all times. Moreover, if $\operatorname{diam}(H) > \theta_{\min}(H)$ and all these interactions are accepted at $t = 0$, then for every sufficiently small $\eta > 0$ some admitted weights make a member refuse, within $2|H|$ steps, an interaction that it accepted at $t = 0$.*

Proof. (a) For fixed v, the function $p \mapsto \|p - x_v\|$ is convex, so its maximum over the convex hull is attained at one of the points x_u. Hence v accepts the whole hull if and only if $\|x_u - x_v\| \leq \theta_v$ for all $u \in H$; requiring this for every v gives the pairwise condition.

(b) The proof of Theorem 1(a) requires only that each update be a convex combination, so it applies to all admitted weights and establishes the forward direction. For the converse, assume $\operatorname{diam}(H) > \theta_{\min}(H)$ and that every interaction along an edge of $G[H]$ is accepted at $t = 0$, since otherwise there is nothing to prove; let $D = \operatorname{diam}(H)$ and $0 < \delta \leq \frac{1}{2}$. In a *copy step* along an edge (z, y), member y puts weight $1 - \delta$ on z and spreads δ evenly over its other accepted neighbours and itself, while every other member puts weight $1 - \delta$ on itself and spreads δ evenly over its accepted neighbours, or keeps weight 1 if it accepts none. Every weight on an accepted neighbour or on the member itself is at least $\delta/|H|$, so these weights are admitted for every $\eta \leq \delta/|H|$, provided that y still accepts z; if it does not, an interaction accepted at $t = 0$ has been refused. The same holds if any member refuses any neighbour during the steps below, so assume that this never happens. By Lemma 1, the diameter stays at most D; hence in a copy step y moves to within δD of the position of z, and every other member moves by at most δD. *Copying along a path* $p_0, \ldots, p_L$, that is, performing copy steps along $(p_0, p_1), \ldots, (p_{L-1}, p_L)$ in this order, therefore brings p_L to within $L\delta D$ of where p_0 started, and moves every member outside $\{p_1, \ldots, p_L\}$ by at most $L\delta D$.

If some pair $a, b \in H$ has $\|x_a - x_b\| > \theta_a$, then a and b are not adjacent. A shortest path from b to a, without its last vertex a, ends at a neighbour y of a and has length $1 \leq L < |H|$. Copying along it brings y to within $L\delta D$ of x_b, while a moves by at most $L\delta D$, so a refuses y at $t = L$ once $2L\delta D < \|x_a - x_b\| - \theta_a$.

Otherwise the pairwise condition holds. Let $\theta_w = \theta_{\min}(H)$ and $\|x_u - x_v\| = D$; then $w \notin \{u, v\}$, because $\|x_w - x_y\| \leq \theta_w < D$ for all $y \in H$. There is a path from u to w that avoids v, or one from v to w that avoids u: if v lies on every path from u to w, then every path from v to w continues inside the component of $G[H] - v$ that contains w, and u is not in this component. Swapping u and v if necessary, copy along a path from u to w that avoids v, and then along a shortest path from v to w without its last vertex w; the latter ends at a neighbour y of w, possibly $y = v$. After these $T < 2|H|$ steps, w is within $T\delta D$ of x_u and y within $T\delta D$ of x_v, so $\|x_w(T) - x_y(T)\| \geq (1 - 2T\delta)D > \theta_w$ once $2T\delta D < D - \theta_w$: member w refuses y, which it accepted at $t = 0$. □

Part (b) is a worst-case statement over all weightings, and its converse requires weights that may be arbitrarily uneven: in the proof, each moving member puts almost all its weight on one neighbour, whereas under equal weights, a group that violates (iii) may or may not lose an accepted interaction (Proposition 1(b)); for instance, let u, w, v form a triangle in G, at positions 0, $1/2$ and 1 on a line, with tolerances 1, 0.6 and 1: every pair satisfies the pairwise condition and (iii) fails, but with $\eta = \frac{1}{3}$ every update is the equal-weight one, so the group meets at its centroid after one step and no interaction is ever refused. The proof requires only a connected interaction graph, so (b) also holds when all members hear one another. With $\theta^{\mathrm{acc}} = \theta^{\mathrm{rej}}$, the same steps are admitted by (2), where a refusal is a rejection, so (iii) is also exactly what rules out rejection under (2) for all coefficients; this needs $\theta^{\mathrm{acc}} = \theta^{\mathrm{rej}}$, since, for instance, with $\theta^{\mathrm{acc}} \equiv 0.4$ in the triangle above every

pair is noncommittal and nobody ever moves. Section VI-C measures how often other methods' groups lose an accepted interaction under equal and uneven weights.

Open groups: When members average with accepted neighbours outside the group, condition (iii) alone no longer suffices; Proposition 3 supplies an additional sufficient condition.

Proposition 3 (Open groups)**.** *Let each $v \in H$ also interact with its outside neighbours $O_v = N_G(v) \setminus H$, whose positions x_o stay fixed, and let every update move x_v to a convex combination, with non-negative weights that may vary with time and state, of $x_v(t)$ and the in-group and outside neighbours that v accepts at time t. Let Y_0 be the set of outside positions accepted by some member at $t = 0$, and $K_0 = \mathrm{conv}(\{x_v(0) : v \in H\} \cup Y_0)$. If (a) $\mathrm{diam}(K_0) \le \theta_{\min}(H)$ and (b) $\|x_o - p\| > \theta_v$ for all $p \in K_0$, $v \in H$ and $o \in O_v$ with $x_o \notin Y_0$, then $x_v(t) \in K_0$ for all v and t, and at every time each member accepts exactly the neighbours it accepted at $t = 0$.*

Proof. We prove by induction on t that $x_u(t) \in K_0$ for all $u \in H$; the claim holds at $t = 0$. Assume it holds at time t and fix v. If $o \in O_v$ and $x_o \notin Y_0$, then $\|x_v(t) - x_o\| > \theta_v$ by (b), so v does not accept o; every other point that v can average, namely $x_v(t)$, the positions $x_u(t)$ of its in-group neighbours and the outside positions in Y_0, lies in the convex set K_0, so $x_v(t+1) \in K_0$. Any two points of K_0 are within $\mathrm{diam}(K_0) \le \theta_{\min}(H) \le \theta_v$, so at every time v accepts all its in-group neighbours and all outside neighbours with positions in Y_0, and no others; by the definition of Y_0, these are exactly the neighbours it accepts at $t = 0$. As a point set and its convex hull have the same diameter (Lemma 1), (a) can be checked on the finite set $\{x_v(0)\}_{v \in H} \cup Y_0$. The argument requires neither connectivity nor positive weights nor synchronous updates, but it does require fixed outside positions. □

The fixed outside neighbours are the constant signals of Hegselmann and Krause [35]; what matters is the certificate. The condition of Mirtabatabaei and Bullo [33, Prop. 4.11], under which an isolated all-to-all group with scalar opinions reaches consensus if all members but one have bounds larger than its opinion range, permits one exemption but no outside neighbours, the reverse of Proposition 3; neither condition therefore implies the other. In our network model, two exemptions are too many by Proposition 1(b). Condition (a) of Proposition 3 alone does not suffice, even with equal weights, because members can begin to accept outside neighbours that none accepted at $t = 0$ and drift apart, an instance of the indirect influence described in [35]. For example, let adjacent members a and b start at 0 and $\frac{1}{2}$ with $\theta \equiv 1$, let a have 20 outside neighbours at 0, and b have 10 at $\frac{19}{20}$ and 10 at $\frac{8}{5}$. Then $\mathrm{diam}(\{x_a, x_b\} \cup Y_0) = \frac{19}{20} \le 1$, but (b) fails; under equal-weight HK, b starts to accept the neighbours at $\frac{8}{5}$ after one step, and at $t = 2$ the two members are $\frac{51}{44} > 1$ apart and refuse each other. The argument needs outside positions to stay fixed, and open groups need not reach consensus: if a has 20 outside neighbours at 0 and b has 10 at 0.95, with $\theta \equiv 1$ and members at 0 and $\frac{1}{2}$, the two members converge to $\frac{19}{460}$ and $\frac{399}{460}$ under equal weights, so the members' diameter grows from $\frac{1}{2}$ to $\frac{19}{23}$ while K_0 stays fixed. The conditions of Proposition 3 rarely hold in our data (Section VI-C).

B. Complexity

We study LCC-DEC in $\mathbb{R}^m$ with the Euclidean norm and rational coordinates and tolerances; in parts (a), (b) and (f), m is part of the input and we use the following embedding.

Lemma 2. *Given a graph $F = (U, E_F)$ with $|U| = n$, points $y_u \in \{-1, 0, 1\}^m$ with $m = O(n^2)$ and an integer r can be computed in polynomial time such that, for distinct $u, w \in U$, $\|y_u - y_w\| \le r$ if and only if $\{u, w\} \in E_F$.*

Proof. Let $\bar{F}$ denote the complement of F, and let Δ denote its maximum degree. Take the least positive integer j satisfying $2j^2 \ge \Delta$, and put $D = 2j^2$. For every edge $\{u, w\}$ in $\bar{F}$ ordered by $u < w$, introduce a coordinate assigning $y_u = +1$, $y_w = -1$, and every other point the value 0. For each u, introduce another $D - \deg_{\bar{F}}(u)$ private coordinates with value 1. This gives $\|y_u\|^2 = D$ for every u; distinct points can share a non-zero coordinate only when they are adjacent in $\bar{F}$. Thus $\langle y_u, y_w \rangle$ is -1 when $\{u, w\} \notin E_F$ and 0 when $\{u, w\} \in E_F$. Thus $\|y_u - y_w\|^2$ takes value $2D+2$ in the former case and $2D$ in the latter, so the threshold $r = 2j$ distinguishes them. □

Theorem 3 (Hardness)**.** (a) *For each fixed $k \ge 1$,* LCC-DEC *is NP-complete. This holds even for complete G, equal tolerances, and coordinates restricted to $\{-1, 0, 1\}$.*

(b) *For every fixed $\varepsilon \in (0, 1)$, unless* P = NP*, no polynomial-time algorithm can approximate* MAXLCC *to within a factor of $n^{1-\varepsilon}$.*

(c) *The conclusions of (a) and (b) extend to general metrics supplied as distance matrices.*

(d) *When $m = 1$,* MAXLCC *can be solved in $O(n^2(n+|E|))$ time.*

(e) *For each fixed $k \ge 1$ and $m \ge 2$,* LCC-DEC *remains NP-complete, also on the sphere with chord distance, even with equal tolerances.*

(f) *For each fixed $k \ge 2$, deciding whether an LCC exists is NP-complete, even with equal tolerances, both for distances in $\{1, 2\}$ and in $\mathbb{R}^m$ with m part of the input. Hence, for each fixed $k \ge 2$, unless* P = NP*, no polynomial-time algorithm returns an LCC for every input that has one; in particular,* MAXLCC *cannot be approximated within any factor.*

Proof. Feasibility can be checked by comparing squared distances and squared tolerances using rational arithmetic, so membership in NP follows directly.

(a) The reduction is from CLIQUE, which is NP-complete [43]. For an instance (F, c), add a clique K of $k+1$ new vertices, including q, and make every vertex of K adjacent to every vertex of U. Apply the embedding of Lemma 2 to this augmented graph, take G to be complete, assign tolerance r to every vertex, and set $s = c + k + 1$. Any c-clique Q in

F gives the set $Q \cup K$, whose pairwise distances are at most r and whose minimum degree in G is $c + k \geq k$. Conversely, every pair of vertices in an LCC has distance at most r; by Lemma 2, these vertices form a clique in the augmented graph, and deleting K leaves a clique in F with at least c vertices.

(b) Embed F itself by Lemma 2, with complete G and tolerance r. Checking all sets of at most $k+1$ vertices in $n^{O(k)}$ time either finds a maximum clique, if its size ω is at most k, or shows $\omega > k$. In the latter case, apply a ρ-approximation algorithm to each query $q \in U$ that lies in a $(k+1)$-clique. If q belongs to a maximum clique, the corresponding optimum equals ω. Every LCC returned by the algorithm is a clique in F, and the largest returned set therefore contains at least ω/ρ vertices. This gives a polynomial-time approximation for CLIQUE within ρ; taking $\rho = n^{1-\varepsilon}$ would imply P = NP [44].

(c) The embedding would give irrational distances, so we bypass it: set $d(u, w) = 1$ if $\{u, w\}$ is an edge of the graph being embedded and $d(u, w) = 2$ for every other pair $u \neq w$, and give every vertex tolerance 1. Any symmetric matrix with zero diagonal and off-diagonal entries in $\{1, 2\}$ satisfies the triangle inequality, and the arguments of (a) and (b) apply unchanged.

(d) Take an optimal solution H^* and define $\tau^* = \theta_{\min}(H^*)$ and $a^* = \min_{v \in H^*} x_v$. The inequality $\operatorname{diam}(H^*) \leq \tau^*$ places H^* inside the window $[a^*, a^* + \tau^*]$. For each pair $(a, \tau) \in \{x_u\} \times \{\theta_u\}$, evaluate $C(a, \tau) = C_k(S_{a,\tau}, q)$, where $S_{a,\tau} = \{v : \theta_v \geq \tau,\ a \leq x_v \leq a + \tau\}$. Every non-empty $C(a, \tau)$ is an LCC: its positions occupy a window of length τ, and each tolerance is at least τ. The k-core fact gives $H^* \subseteq C(a^*, \tau^*)$, establishing optimality of the largest $C(a, \tau)$. The enumeration considers n^2 pairs and takes $O(n + |E|)$ time for each.

(e) We reduce from 3-SAT [43] with variables $x_1, \ldots, x_\nu$, $\nu \geq 2$, and clauses $c_1, \ldots, c_\mu$. Let $t_i = i/(\nu + 1)$ and $p_i = \big((1 - t_i^2)/(1 + t_i^2),\ 2t_i/(1 + t_i^2)\big)$, a rational point of the unit circle at angle $\phi_i = 2 \arctan t_i \in (0, \pi/2)$. Place a clique K of $k + 1$ vertices, including q, at the origin; for each i, place k *copies* of the literal x_i at p_i and k copies of $\neg x_i$ at $-p_i$; and place one vertex per clause at the origin. Every copy is adjacent to all of K, and each clause vertex is adjacent to all copies of its literals. Copies of x_i and $\neg x_i$ are at distance 2; every other pair is closer than $r = 2 - 1/(16(\nu+1)^2)$. Indeed, pairs involving the origin and pairs of copies of the same literal are at distance at most 1. Copies of literals of distinct variables x_i and x_j are at distance $2\sin(\alpha/2)$ or $2\cos(\alpha/2)$, where $\alpha = |\phi_i - \phi_j| < \pi/2$; both are at most $2\cos(\delta/2)$, where $\delta \leq \alpha$ is the smallest difference between distinct angles ϕ_i. Since the derivative of $2 \arctan t$ is at least 1 on $[0, 1]$ and consecutive values t_i differ by $1/(\nu + 1)$, we have $\delta \geq 1/(\nu + 1)$; with $y = \delta/2 < 1$ and $\cos y \leq 1 - y^2/2 + y^4/24 \leq 1 - 11y^2/24$, this gives $2\cos(\delta/2) \leq 2 - 11\delta^2/48 < r$. With tolerance r for every vertex, an LCC contains copies of at most one literal per variable, and therefore at most $s = k + 1 + k\nu + \mu$ vertices. An LCC with s vertices contains K, all copies of one literal per variable, and every clause vertex; each clause vertex then needs k neighbours, that is, a chosen literal, so the chosen literals satisfy the formula. Conversely, the true literals of a satisfying assignment, together with K and all clause vertices, form an LCC with s vertices. On the sphere with chord distance, put the copies on the equator and the other vertices at the north pole, at distance $\sqrt{2} < r$ from every copy; the argument is unchanged. Padding the coordinates with zeros extends the planar construction to every fixed $m \geq 2$.

(f) Membership in NP is as above. We reduce from MULTI-COLORED CLIQUE: given a graph $F = (U, E_F)$ whose vertex set is partitioned into classes $U_1, \ldots, U_m$ with $m \geq 2$, decide whether some $u_1 \in U_1, \ldots, u_m \in U_m$ are pairwise adjacent. It is NP-complete, since an instance (F', c) of CLIQUE with $c \geq 2$ reduces to it by taking $U_i = V(F') \times \{i\}$ for $i \leq c$ and joining (a, i) and (b, j) if and only if $i \neq j$ and $\{a, b\} \in E(F')$. Add an original vertex $\hat{q}$ and arrange the classes $\{\hat{q}\}, U_1, \ldots, U_m$ on a ring, so that the *ring neighbours* of U_i are U_{i-1} and U_{i+1}, with $U_0 = U_{m+1} = \{\hat{q}\}$. Replace every original $a \in U \cup \{\hat{q}\}$ by $k - 1$ copies, let q be a copy of $\hat{q}$, and join two copies in G if and only if their originals lie in ring-neighbouring classes. Call two originals *compatible* if they are equal, if one of them is $\hat{q}$, or if they lie in different classes and are adjacent in F. Put copies of compatible originals at distance 1 and all other pairs at distance 2, with tolerance 1 for every vertex; for Euclidean distances, embed the compatibility graph on $U \cup \{\hat{q}\}$ by Lemma 2, put every copy at the point of its original, and give every vertex tolerance r. In both cases, a set satisfies (iii) if and only if its members are copies of pairwise compatible originals. If $u_1, \ldots, u_m$ form a multicolored clique, all copies of $\hat{q}, u_1, \ldots, u_m$ form an LCC: the subgraph they induce is connected, each of its vertices has $2(k - 1) \geq k$ neighbours in it, and their originals are pairwise compatible. Conversely, let H be an LCC. Distinct originals in the same class are incompatible, so the members of H among the copies of a class are copies of a single original. The neighbours in $G[H]$ of a member therefore lie in its two ring-neighbouring classes, at most $k - 1$ in each, and minimum degree k forces both of these classes to contribute members. Starting from q, whose ring neighbours are U_1 and U_m, and following the ring, every class U_i contributes copies of some u_i; these originals are pairwise compatible and lie in different classes, so they form a multicolored clique. Finally, if a polynomial-time algorithm returned an LCC for every input that has one, checking whether its output is an LCC would decide existence in polynomial time. □

Parts (a) and (b) transfer the hardness of CLIQUE through Lemma 2, a simple instance of representing graphs by distances as in the study of sphericity [45], and (c) through distances in $\{1, 2\}$; (e) and (f) require new constructions. With uniform tolerances, (a) and (e) also show that the query version of the (k, r)-core problem [3] is NP-hard under Euclidean distance, even in the plane, and by (f), for each fixed $k \geq 2$, even deciding whether q lies in any (k, r)-core is NP-complete when the dimension is part of the input. The assumption that k is fixed matters: if k is part of the input, the construction of (a) with $K = \{q\}$ and $k = c$ already makes existence NP-

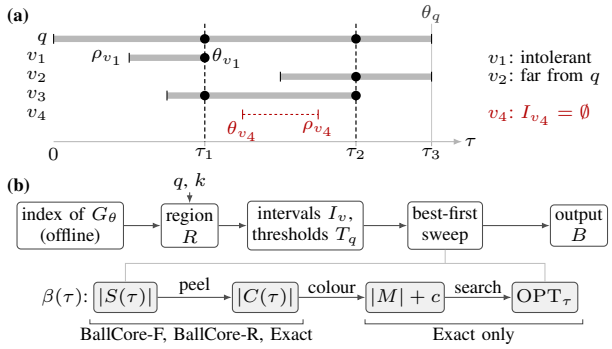


Fig. 2. The LatSweep framework: (a) survival intervals and the candidates $S(\tau)$ (dots) at two thresholds; (b) index, query pipeline and the bounds $\beta(\tau)$ that the sweep refines.

complete. Parts (b) and (f) motivate relaxing the diameter, but not the size, in Section V-B: by (f), unless P = NP, every polynomial-time algorithm that returns only LCCs, such as BallCore-F and the greedy variant, returns nothing on some inputs that have an LCC, whereas BallCore-R is empty only if no LCC exists. Part (b) is not subsumed by (f): it also holds for $k = 1$, and its proof gives the same bound even on inputs with an LCC that is easy to find, a $(k+1)$-clique of F containing q.

Open cases: In fixed dimension, it remains open whether MaxLCC is hard to approximate and whether, for $k \geq 2$, deciding the existence of an LCC is hard; on the sphere with chord distance, as in our meetup data, only (e) applies. If G is complete, Lemma 4 reduces MaxLCC to finding, for each candidate threshold, a maximum clique containing q in a unit ball graph. This is polynomial in the plane [46], even with member-specific tolerances, so the hardness in (e) stems from the graph; in dimension 4 and above, it is NP-hard even for complete G [47], while these results leave dimension 3 unresolved.

V. The LatSweep Framework

LatSweep makes the coupled constraint (iii) searchable by fixing the smallest tolerance τ. Each value yields a fixed-threshold subproblem in a reduced region around q (Fig. 2). The sweep yields four algorithms: BallCore-F and a greedy variant return only LCCs, BallCore-R is a polynomial-time $(1, 2)$-bicriteria approximation, and LatSweep-Exact (*Exact*) is exact and, if stopped early, returns the best LCC found, if any, with an upper bound on the optimum. For a fixed threshold, the subproblem is a search for a maximum (k, r)-core containing q [3], which we solve with known tools: peeling, filtering of conflicts (pairs farther apart than the threshold) and colouring bounds [48]. Survival intervals provide the reduction despite non-monotone candidate sets. Best-first refinement then allocates search across thresholds and supports early stopping with a certificate; a compatibility index reduces the search region. We write $\rho_v = d(x_v, x_q)$ and let OPT be the size of a largest LCC, or 0 if none exists; all results hold in arbitrary metric spaces.

A. Survival Intervals and Candidate Reduction

Lemma 3 (Sound reductions)**.** *Let H be an LCC and $\tau = \theta_{\min}(H)$. (a) Every $v \in H$ satisfies $\rho_v \leq \tau \leq \theta_v$, and hence $\rho_v \leq \min(\theta_v, \theta_q)$. (b) Let $G_\theta = (V, E_\theta)$ with $E_\theta = \{(u, v) \in E : d(x_u, x_v) \leq \min(\theta_u, \theta_v)\}$. Then $G[H] = G_\theta[H]$; hence H lies in the component of the k-core of G_θ that contains q.*

Proof. (a) As $q \in H$, we have $\rho_v \leq \mathrm{diam}(H) \leq \theta_{\min}(H) = \tau \leq \theta_v$, and $\tau \leq \theta_q$. (b) Every edge (u, v) inside H satisfies $d(x_u, x_v) \leq \mathrm{diam}(H) \leq \theta_{\min}(H) \leq \min(\theta_u, \theta_v)$; the k-core fact applied to G_θ completes the proof. □

Lemma 3(a) assigns each vertex v a *survival interval* $I_v = [\rho_v, \min(\theta_v, \theta_q)]$ for the smallest tolerance of an LCC containing it. A vertex with an empty interval, such as v_4 in Fig. 2(a), cannot belong to an LCC. The core numbers of G_θ form a *compatibility index*; it is independent of the query and is built once per tolerance assignment (say, for θ^{acc} and for θ^{rej}) in $O(t_d|E|+n)$ time, where t_d is the cost of one distance evaluation. For a query and a graph $G' \in \{G, G_\theta\}$, the *query region* R is obtained by a traversal from q along edges of G' that visits only vertices with a non-empty survival interval and core number at least k in G'. By Lemma 3, every LCC lies in R for either choice of G', and building R inspects only R and its neighbours; BallCore-R uses $G' = G$, and hence the core numbers of G rather than the index, all other variants $G' = G_\theta$.

Candidate thresholds: For $\tau > 0$, let $S(\tau) = \{v \in R : \tau \in I_v\}$ and let $C(\tau) = C_k(S(\tau), q)$, peeled in G'. Let $T_q = \{\min(\theta_v, \theta_q) : v \in R\}$, and let OPT_τ be the largest size of a connected set $H \subseteq S(\tau)$ with $q \in H$, minimum degree at least k and $\mathrm{diam}(H) \leq \tau$.

Lemma 4. $\mathrm{OPT} = \max_{\tau \in T_q} \mathrm{OPT}_\tau$*, and* $\mathrm{OPT}_\tau \leq |C(\tau)| \leq |S(\tau)|$ *for all* τ.

Proof. Every set counted in OPT_τ has all tolerances at least $\tau \geq \mathrm{diam}(H)$ and is therefore an LCC. Conversely, an optimal H^* has $\tau^* = \theta_{\min}(H^*) \leq \theta_q$, so $\tau^* \in T_q$, and $H^* \subseteq S(\tau^*)$ by Lemma 3(a); thus H^* is counted in OPT_{τ^*}. The bound $\mathrm{OPT}_\tau \leq |C(\tau)|$ is the k-core fact. □

Best-first sweep: Algorithm 1 keeps an upper bound $\beta(\tau)$ on OPT_τ for every threshold and repeatedly refines the bound of the threshold with the largest one, in stages of increasing cost (Fig. 2(b)): the interval count $|S(\tau)|$, obtained for all thresholds by one sort of the interval end points; the size $|C(\tau)|$ of the peeled set; for Exact, the bound that branch and bound computes at its root (Section V-C); and finally the search itself, which settles OPT_τ. The root bound holds only for solutions larger than the incumbent B, which suffices because B never shrinks. By Lemma 4, no threshold can improve the incumbent once every remaining bound is at most $|B|$. Candidate count alone can prioritize expensive subproblems with small optima. The current bound incorporates the refinements beyond $|S(\tau)|$; the ablation in Section VI-B tests the benefit of this ordering. Unlike with a uniform radius, $S(\tau)$ is not monotone in τ: a larger τ admits vertices farther from q

Algorithm 1 LatSweep

Require: G, x, θ, query q, integer k, variant $\in \{\mathsf{F}, \mathsf{R}, \mathsf{Exact}\}$

$\lambda \leftarrow 2$ for F, else 1; $G' \leftarrow G$ for R, else G_θ
$R \leftarrow$ query region in G', with $\lambda\rho_v$ for ρ_v ▷ Lemma 3
$I_v \leftarrow [\lambda\rho_v, \min(\theta_v, \theta_q)]$ for $v \in R$
$T \leftarrow \{\min(\theta_v, \theta_q) : v \in R\}$; $\beta(\tau) \leftarrow |S(\tau)|$ for $\tau \in T$
$B \leftarrow \emptyset$, or for Exact the larger of the BallCore-F and greedy results
while $T \neq \emptyset$ and $\max_{\tau \in T} \beta(\tau) > |B|$ **do**
  $\tau \leftarrow$ a threshold in T with the largest $\beta(\tau)$
  if τ has not been peeled **then**
    $C(\tau) \leftarrow C_k(S(\tau), q)$ in G'; $\beta(\tau) \leftarrow |C(\tau)|$
    if the variant is F or R **then**
      if $|C(\tau)| > |B|$ **then** $B \leftarrow C(\tau)$
      remove τ from T
  else if τ has no root bound yet **then**
    $\beta(\tau) \leftarrow$ root bound of $C(\tau)$ ▷ Section V-C
  else
    $H \leftarrow$ BranchAndBound$(C(\tau), \tau, |B|)$
    if $|H| > |B|$ **then** $B \leftarrow H$
    remove τ from T ▷ the threshold is settled
return B

but drops those with $\theta_v < \tau$; in Fig. 2(a), from τ_1 to τ_2, $S(\tau)$ loses v_1 and gains v_2. The sort takes $O(n_q \log n_q)$ time and one peeling $O(n_q + e_q)$, where $n_q = |R|$ and $e_q = |E(R)|$.

B. BallCore: Feasible and Bicriteria Solutions

Feasible solutions: Variant F doubles the left end points of the survival intervals ($\lambda = 2$ in Algorithm 1); it is fast but conservative, since it admits only vertices within $\tau/2$ of q. The greedy variant starts from $C(\tau)$, peeled in G_θ, at each of the 20 thresholds with the largest $|S(\tau)|$. It repeatedly deletes the vertex other than q with the most *conflicts*, that is, partners at distance greater than τ, and re-peels after each deletion until no conflict remains; if peeling removes q, the threshold yields nothing.

Proposition 4 (Feasibility)**.** *Every non-empty output of BallCore-F or of the greedy variant is an LCC.*

Proof. By peeling, each output contains q, is connected, and has minimum degree at least k and tolerances at least τ. In BallCore-F, any two members satisfy $d(x_u, x_v) \leq \rho_u + \rho_v \leq \tau/2 + \tau/2 = \tau$; in the greedy output, no pair conflicts. □

BallCore-R: Variant R, for *relaxed*, omits the doubling and uses the largest peeled set as a bicriteria approximation. It peels in $G' = G$, so that $C(\tau)$ is the largest connected k-core containing q whose members lie within distance τ of q and have tolerance at least τ. Since $S(\tau)$ can only grow between consecutive values of T_q, maximizing over $\tau \in T_q$ solves exactly the relaxation of MaxLCC that replaces (iii) by the requirement that $\rho_v \leq \tau \leq \theta_v$ hold for all members and some τ; every LCC satisfies this relaxation by Lemma 3(a).

Theorem 4 (Bicriteria approximation)**.** *Let $\hat{\tau}$ maximize $|C(\tau)|$ over $\tau \in T_q$, and let $B = C(\hat{\tau})$. Then $|B| \geq$ OPT; in particular, B is empty only if no LCC exists. If B is non-empty, it contains q, is connected, has minimum degree at least k, and satisfies $\mathrm{diam}(B) \leq 2\,\theta_{\min}(B)$. Given the region, BallCore-R finds B in $O(n_q \log n_q + |T_q|(n_q + e_q))$ time.*

Proof. By the proof of Lemma 4 and the k-core fact, an optimal H^* satisfies $H^* \subseteq C(\tau^*)$ for some $\tau^* \in T_q$, so $|B| \geq |C(\tau^*)| \geq$ OPT. For any $u, v \in C(\tau)$, we have $d(x_u, x_v) \leq \rho_u + \rho_v \leq 2\tau$, and every tolerance in $C(\tau)$ is at least τ; hence $\mathrm{diam}(C(\tau)) \leq 2\tau \leq 2\,\theta_{\min}(C(\tau))$. Algorithm 1 stops only when every remaining threshold has $|S(\tau)| \leq |B|$ and therefore cannot improve B. □

The factor 2 is attained: place q at 0 and k vertices at each of $-\tau$ and $+\tau$ on a line, take G complete, and give every vertex tolerance τ. Then $C(\tau)$ contains all $2k + 1$ vertices and has diameter 2τ, whereas an optimum consists of q and the k vertices on one side. It is also best possible in general metrics: with distances in $\{1, 2\}$ as in Theorem 3(c), every set of diameter below 2 is a clique, so unless P = NP, no polynomial-time algorithm achieves size ratio $n^{1-\varepsilon}$ with diameter factor $2 - \delta$. The role of BallCore-R is to bound OPT and certify absence when its output is empty. These uses require a completed sweep, and its output has none of the dynamical guarantees of Section IV.

C. LatSweep-Exact

By Lemma 4, an optimal LCC is found by solving, at every $\tau \in T_q$ whose bound exceeds the incumbent, the fixed-threshold subproblem on $C(\tau)$: two vertices *conflict* if their distance exceeds τ, and the goal is a largest conflict-free subset containing q that is connected and has minimum degree at least k. The exact variant peels in G_θ, which is sparser and by Lemma 3(b) loses no solution, and initializes the incumbent with the BallCore-F and greedy solutions, which are feasible by Proposition 4.

Branch and bound: BranchAndBound(C, τ, b) returns a largest feasible subset of C if it has more than b vertices, and the empty set otherwise. A search node (M, P) consists of selected vertices $M \ni q$ and undecided vertices P, with no conflicts within M or between M and P; the root has $M = \{q\}$ and $P = C(\tau) \setminus \{q\}$, which satisfies this invariant because $\rho_v \leq \tau$ for every $v \in C(\tau)$. Let b denote the size of the incumbent. Each node first applies rules (1) and (2) of Lemma 5 to reduce $M \cup P$. If no conflict remains in P, then $M \cup P$ is feasible and optimal for the subtree; otherwise, the node tests rule (3) and branches on a vertex $p \in P$ with the most conflicts in P, either deleting p or moving it to M and deleting its conflicting partners from P. Together, the two branches retain every solution in the subtree, and both maintain the invariant. The *root bound* used by Algorithm 1 is $|M| + c$ at the root after rules (1) and (2), where $c \leq |P|$ is the colour count of rule (3); by Lemma 5, no feasible subset of $C(\tau)$ with more than b vertices exceeds it. Rules (2) and (3) adapt the degree and colouring bounds of maximum-clique search [48].

Lemma 5 (Sound pruning)**.** *The rules below preserve every feasible solution with more than b vertices in the current node's subtree. (1) Delete edges joining conflicting vertices, peel $G[M \cup P]$ down to the component containing q in its k-core, and prune the node whenever this removes a vertex of M. (2) Prune the node when $|M \cup P| \leq b$. If this test fails, examine any vertex having fewer than b conflict-free partners in $M \cup P$: delete it if it belongs to P, or prune the node if it belongs to M. Apply rules (1) and (2) alternately until both leave $M \cup P$ unchanged. (3) Greedily colour the vertices of P, in increasing order of their number of conflicts in P, so that every colour class is a set of pairwise conflicting vertices; prune when $|M|$ plus the colour count c does not exceed b.*

Proof. Take any feasible H satisfying $|H| > b$ and $M \subseteq H \subseteq M \cup P$. (1) Since H contains no conflicting pair, none of the deleted edges belongs to $G[H]$; the k-core fact therefore ensures that H remains after peeling. (2) Every $v \in H$ has the remaining $|H| - 1 \geq b$ vertices of H as conflict-free partners, and all are contained in $M \cup P$. (3) $H \setminus M$ is pairwise conflict-free, so it can contain at most one vertex from each colour class. □

Colouring vertices with few conflicts first tends to use fewer colours than an arbitrary order [48] and gives a smaller additional speed-up in our experiments. Conflicts are stored as bitsets over $C(\tau)$, which takes $O(|C(\tau)|^2)$ time and space per threshold. The reductions shrink the region by orders of magnitude for most evaluated queries (Section VI). The remaining search is still exponential in the worst case, as expected from Theorem 3.

Time limits and certificates: Our implementation checks a time limit at every search node and does not search sets $C(\tau)$ with more than 3,000 vertices; such thresholds keep their bound, and root bounds are computed only for sets of up to 20,000 vertices, which Greedy also skips if larger. When it stops before every threshold is settled, a pass outside the time limit, of at most 5 s or 10% of the limit, whichever is larger, tightens the bounds of the unsettled thresholds T; it then returns the incumbent B with $\mathrm{UB} = \max(|B|, \max_{\tau \in T} \beta(\tau)) \geq \mathrm{OPT}$. Every answer is thus proven optimal ($\mathrm{UB} = |B|$, which includes proven empty answers), feasible with a gap $|B|/\mathrm{UB}$ ($0 < |B| < \mathrm{UB}$), or undecided ($B = \emptyset$ and $\mathrm{UB} > 0$).

VI. Experiments

The experiments test search efficiency and solver effectiveness (Section VI-B), persistent acceptance in simulated dynamics (Section VI-C), and the effects of personal tolerances in applications with observed positions: meetups (Section VI-D) and two parliaments (Section VI-E). The parliament analyses follow plans specified before results were available. The size-controlled Gowalla and Brightkite comparisons are post hoc. Shares are compared in percentage points (pp); brackets denote 95% bootstrap confidence intervals (CIs). Between-method CIs resample queries 2,000 times with the two groups paired within each query. Groups may overlap across queries; the CIs therefore describe the sampled queries within each graph. Appendix A gives the full tables of all experiments.

A. Setup

Graphs and positions: Gowalla and Brightkite [49] provide friendships and time-stamped check-ins, which we split by time into a training (70%), a validation (10%) and a test period (20%). We retain users with at least 10 training check-ins and the friendships between them. The home x_v of a user is the median check-in position in the user's most frequent cell of about 25 km, and θ_v is the median distance of the user's training check-ins from home, at least 1 km. Distances are chord lengths between Earth-centred coordinates, within 0.1% of great-circle distances below 300 km. DBLP, YouTube, LiveJournal, Orkut and Friendster [50], together with Pokec [51], lack positions. For these SNAP graphs [52], we generate community-aligned positions. Label propagation [53] with a size cap forms groups of at most 1,000 vertices, whose members scatter with unit Gaussian spread in $\mathbb{R}^m$ around centres about eight times the within-group scale $\sqrt{2m}$ apart. Tolerances are $\theta_v = s\sqrt{2m}\exp(\sigma Z_v - \sigma^2/2)$ with standard normal Z_v and $\sigma = 0.4$, where s is the smallest scale, on a grid refined by bisection, at which Exact (5 s limit per query) finds LCCs with a median of at least 10 members for 30 calibration queries, or the largest scale of the grid if none does (YouTube, and Friendster on a coarser grid); s ranges from 1.25 on Orkut to 3 on YouTube. Synthetic positions test efficiency and dynamics; the application analyses use observed positions.

Queries and algorithms: Efficiency queries are drawn at random from the k-core of G (on synthetic positions, from groups of at least 20 members), 50 per setting. The default setting ($k = 3$, all vertices, $m = 16$ and the calibrated scale) occurs in every parameter sweep on Gowalla, Brightkite, LiveJournal and Orkut, whose disjoint samples we pool into 200 queries per graph (150 on Gowalla and Brightkite, which have no dimension sweep); the other four graphs use only the default setting, with 200 queries each. We run Exact (with a time limit of 60 s), the greedy variant (*Greedy*), BallCore-F and BallCore-R. The connected k-core component of q and a greedy (k, r)-core search with the median tolerance as radius serve as references; on Friendster, the former was timed in a separate run of the same code on the same queries. All algorithms we time are written in C++ (g++ 11.4, `-O3 -march=native`) and run single-threaded on an Intel Xeon Gold 6530 with 64 GB of memory; a two-pass loader converts Friendster's edge list in 13 minutes. Query times exclude loading the graph, building the index and the bound pass after a time limit (Section V-C), which took at most 4 µs on our efficiency queries.

Baselines: The baselines separate the effects of the LCC constraint, personal tolerances and structural cohesion; each dynamics or application analysis uses the relevant subset. The largest group satisfying the pairwise condition (*pairwise*) is NP-hard to find by the proof of Theorem 3(a), since the condition coincides with (iii) under equal tolerances; our

branch and bound solves it on the fixed conflict graph in which u and v conflict if $d(x_u, x_v) > \min(\theta_u, \theta_v)$, with a limit of 60 s (10 s in the heterogeneity sweep, where Exact keeps 60 s). *Edge-accept* is the k-core on the edges accepted by both endpoints. The baselines that ignore tolerances are (k, r)-cores at the LCC's own threshold τ (an oracle baseline, as τ is known only after the search) and at the median tolerance (*global*); SAC [6] in its AppInc approximation, the connected k-core containing q in the smallest ball centred at q that holds one, and *size-matched SAC*, the k-core in the smallest such ball whose k-core has at least the LCC's size, which may therefore be larger; a random connected group of the LCC's size; and the k-core component.

B. Efficiency

Queries with and without an LCC: At the default setting, most queries without an LCC are settled before any search: the reductions in Lemma 3 eliminate every candidate for 74–100% of them, and Exact certifies absence in a median of at most 4.7 µs and a maximum of 0.81 s. Between 13% (Brightkite) and 57% (DBLP) of the queries have an LCC, and Table II reports them separately: Exact takes a median of 0.07 ms (Brightkite) to 861 ms (Pokec), and it proves optimality on all but 4 queries on Pokec and 3 on Friendster, which reach the 60 s limit (column Unproven). Extracting q's k-core component, which ignores positions and tolerances, serves as a reference (median time in Table II): on the seven graphs where it ran alongside Exact, Exact is faster on 70% (Pokec) to 99% (Brightkite) of the default queries and on 41–97% of those with an LCC; on Friendster, it was timed in a separate run (median 70.5 s). For queries with an LCC, Greedy finds one on 77–100%. On queries with a proven optimum, its mean size relative to the optimum is 0.62–0.98, assigning 0 when it finds none. On LiveJournal, Orkut and Pokec, BallCore-F finds an LCC for only 4–6% of queries that have one. BallCore-R is evaluated as an upper bound because its output need not be an LCC. On the four largest graphs, at most 3% of its non-empty outputs satisfy (iii), and its sets have a median size of 1.3–2.0 times the optimum; the greedy (k, r)-core search returns no LCC there.

Unproven queries: Of the 3,700 queries across all graphs and settings, 9 remain unproven by Exact: 2 among the 2,900 parameter-sweep queries and 7 among the 800 queries on the other four graphs. Each unproven query has a returned LCC of at least 0.689 times a proven upper bound on OPT (Section V-C): 0.955 and 0.982 on LiveJournal and Orkut, and 0.689–0.939 on Pokec and Friendster. These queries have 560–918 candidate vertices, and every efficiency query has at most 1,502; the size caps of Section V-C therefore never apply (Table XIV).

Scalability: On Friendster, with 65.6 million vertices and 1.8 billion edges, Exact takes 1.9 s on average, with a median of 0.9 ms and a 90th percentile of 2.0 s, and it proves 98.5% of the queries; on the queries with an LCC, its median is 352 ms and its 90th percentile (p90 in Table II) 7.9 s, whereas extracting q's k-core component takes a median of 70.5 s. The hardest graph is Pokec (median 861 ms, p90 21.6 s for queries with an LCC). The graph G_θ keeps 10–61% of the edges; the index is built once per tolerance assignment in 5 ms (Brightkite) to 9.7 s (Orkut), and in 4.7 minutes on Friendster.

Ablation and parameters: On the 50 queries of the $k = 3$ sample on LiveJournal and Orkut, Exact solves every query within the limit; sweeping in non-increasing order of $|S(\tau)|$ instead of best-first (*count order*) leaves 6% and 8% unproven, colouring in index order instead of by conflicts 2% each, and dropping rule (3) of Lemma 5 12% each. Without the index, the median query region on LiveJournal grows from 7 to 29 vertices and the median time from 0.07 to 0.48 ms; without any reductions or pruning, every query processes q's entire k-core component (2.7 and 3.0 million vertices), and 12% and 14% of the queries reach the limit. Across $k \in \{2, \ldots, 5\}$, vertex samples of 25–100% and $m \in \{2, 4, 8, 16, 32\}$ dimensions, the median time stays below 0.5 ms on LiveJournal and 19 ms on Orkut, and no query reaches the limit. The tolerance scale s matters most: at half the calibrated s, almost no LCC exists (0% and 4% of the queries); at twice and three times the calibrated s, most queries have one (68–88%), and Exact leaves one query unproven on each graph at twice the calibrated s.

Solver comparison: In a common Python framework with the count-order sweep, and with only the solver of the fixed-threshold subproblem exchanged, our branch and bound proves 96.5% of 200 default queries per graph on LiveJournal and Orkut, against 72.5–90.5% for the other solvers (Table III, which also gives the shares among the queries with an LCC found by some solver), and it proves every query that any of them proves; all solvers agree on every proven optimum. We also fix the reductions, the cap on searched sets, the time limit of 60 s and the initial incumbent; the alternatives are our reimplementation of AdvMax, the maximum (k, r)-core algorithm of Zhang et al. [3], adapted to contain q and run with its own and with a simple branching order; a mixed-integer program with flow-based connectivity solved by HiGHS [54]; and the same model in CP-SAT [55]. All solvers, including a Python reimplementation of our branch and bound, run single-threaded; 150 of the queries also appear in Table II, and only relative times are meaningful. On hard queries taking our solver at least 100 ms and proven by both solvers (12–28 per pair), other solvers' median slowdown factors are 5.7 to 175; on queries of 1–100 ms, the AdvMax reimplementations match our branch and bound (median ratios 0.93–1.05), and below 1 ms the others are about twice as fast, a difference of microseconds.

Against a single threshold: Once the index is built, sweeping personal tolerances costs no more per query than the usual single-threshold search: on the 200 queries of Table III, the maximum (k, r)-core containing q, computed by our engine at the median tolerance, times out on 71 queries on Orkut and 19 on LiveJournal, where the sweep proves all 200; at the LCC's own τ^*, run on the 115 and 95 queries with an LCC, it times out on 37 and 16. For queries with an LCC, the sweep is faster on 92–94% on Orkut (median

TABLE II
EXACT AND GREEDY AT THE DEFAULT SETTING ($k = 3$; TIMES IN MS UNLESS MARKED).

				No LCC	LCC exists			Greedy		
Graph	$\|V\|$	$\|E\|$	With LCC	Median (µs)	Median	p90	Unproven	Finds	Size	k-core
Gowalla	56 K	222 K	16%	0.3	0.26	27	0	100%	0.98	2.2
Brightkite	24 K	108 K	13%	0.2	0.07	0.41	0	95%	0.95	1.1
DBLP	317 K	1.0 M	57%	0.9	0.14	5.4	0	96%	0.94	22
YouTube	1.1 M	3.0 M	23%	2.3	31	349	0	93%	0.90	41
Pokec	1.6 M	22.3 M	52%	3.3	861	21.6 s	4	77%	0.62	373
LiveJournal	4.0 M	34.7 M	50%	3.4	6.0	1.9 s	0	90%	0.79	630
Orkut	3.1 M	117.2 M	55%	4.7	200	3.8 s	0	84%	0.68	1.6 s
Friendster	65.6 M	1.81 B	55%	0.4	352	7.9 s	3	89%	0.77	70.5 s

TABLE III
SOLVERS IN THE COUNT-ORDER SWEEP: QUERIES PROVEN OPTIMAL (%) AND MEDIAN SLOWDOWN AGAINST OURS ON HARD QUERIES.

	Proven		Proven, LCC		Slowdown	
Solver	LJ	Ork.	LJ	Ork.	LJ	Ork.
Branch and bound (ours)	96.5	96.5	92.6	93.9	1	1
AdvMax, simple branching	90.5	80.5	80.0	66.1	5.7	7.2
AdvMax	89.5	76.5	77.9	59.1	14.0	14.6
MILP (HiGHS)	90.5	80.0	80.0	65.2	26.7	21.2
CP-SAT	86.0	72.5	70.5	52.2	175	45.0

TABLE IV
GROUPS (%) THAT LOSE AN INTERACTION ACCEPTED AT THE START, ON $G[H]$ (G, B, L, O: GOWALLA, BRIGHTKITE, LIVEJOURNAL, ORKUT).

	Equal				Concentrated				Gossip			
Method	G	B	L	O	G	B	L	O	G	B	L	O
LCC (Exact)	0	0	0	0	0	0	0	0	0	0	0	0
Pairwise	0	0	0	0	40	33	32	40	17	11	3	7
(k, r)-core at τ	15	17	4	13	41	44	91	97	34	40	69	85
Size-matched SAC	5	8	1	2	52	56	89	98	39	40	63	86

time ratios 0.065 and 0.135), and on LiveJournal, where it answers most of them in under 1 ms, the two searches take similar times (median ratios 0.86–0.89 with a 1 ms floor); query times again exclude the index, which only the sweep needs and which would dominate on LiveJournal if amortized over 200 queries (Table XIII). At one threshold, the candidate region ignores the personal tolerances, and the single-threshold answers satisfy (iii) on at most 4.2% of the queries.

Heterogeneity: With equal tolerances ($\sigma = 0$), the LCC coincides with the (k, r)-core at the same threshold on all 220 queries of LiveJournal and Orkut on which Exact and the exact (k, r)-core and pairwise searches finish; with one threshold to sweep, Exact (count order) reaches its limit on 0.5% and 45% of the 200 queries per graph, against 14% and 26.5% at $\sigma = 0.2$ and 4% and 7% at $\sigma = 0.4$ (Table X). From $\sigma = 0.2$ on, (k, r)-cores at the LCC's threshold satisfy (iii) on at most 9% of the queries, and 12–65% of their members face another member beyond their own tolerance.

Tolerance scale and the proven share: Exact's proven share drops once tolerances let LCCs span several communities. The synthetic scales were calibrated to yield LCCs with a median of at least 10 members. Most friends lie in different communities, so tolerances of $c \geq 0.75$ (Orkut) or $c \geq 1$ (LiveJournal) times the median distance between friends let LCCs span several communities. In these settings, Exact reaches the limit on 35–99% of the queries with an LCC (0–1.7% stop at a cap), with loose bounds (median solution-to-bound ratios of 0.03–7.6%); at $c = 1$, it proves only 64% (LiveJournal) and 20% (Orkut) of them, and for $c = 0.5$, 0.75, 1, 1.5 and 2, 100%, 100%, 64%, 15% and 3% on LiveJournal and 100%, 51%, 20%, 3% and 0.6% on Orkut (Table XI). On queries drawn from the whole k-core, it proves all. Cap and time-limit ablations test why the proven share drops (Section V-C): at $c = 1$ on both graphs and $c = 0.75$ on Orkut, raising the cap on searched sets to 20,000 or 50,000 or the root-bound cap to 100,000 changes the proven share by at most $+1.1$ pp, letting Greedy use all thresholds lowers it by up to 3.3 pp, and a limit of 300 s proves only 0–9% of the remaining queries (Table XII). On Orkut, the queries with an LCC have a median of 8,360 candidate vertices at $c = 0.75$ and 210,146 at $c = 1$, whereas no efficiency query has more than 1,502; on LiveJournal at $c = 1$, the median is 1,218 (Section A-E).

C. Dynamics on Returned Groups

We run the dynamics of Section IV on the groups returned by each method for queries with an LCC (a random 1,000 of the meetup queries of Section VI-D on Gowalla, all 462 on Brightkite, and the 95 and 115 default queries of Table III on LiveJournal and Orkut; for pairwise, also the 3, 2, 12 and 14 groups of queries without an LCC). Interactions follow the edges of $G[H]$ for at most 300 steps. We test whether any initially accepted interaction is later refused. In addition to equal-weight HK (1), the variants in Table IV draw each member's weights from a Dirichlet distribution with $\alpha = 0.1$, which concentrates them on a few neighbours, or let random adjacent pairs move towards each other (Deffuant–Weisbuch gossip [32]); randomized variants run 10 times, and a group counts if any run loses an interaction.

Closed groups: No group satisfying (iii) loses an accepted interaction in 83,594 group–variant pairs across 11

variants, including complete interaction graphs, as Theorem 1 predicts. Under SJT dynamics (2), none of 15,392 shows a new rejection. The zero counts check the implementation against the theorem; the other rows of Table IV compare groups without its guarantee. Even with equal weights, groups found without tolerances lose accepted interactions ((k, r)-cores at τ: 4–17%); this occurs much more often with uneven influence (34–98%). With equal weights, pairwise groups never lose an accepted interaction, although 62–98% violate (iii) and fall outside Theorem 1. Under gossip, 3–17% lose one, as do 32–40% under concentrated weights. Usually a single pair is affected (median refused-interaction share at most 0.3%). On Gowalla and Brightkite with concentrated weights, refusal occurs mostly in the first step, and acceptance resumes in about 99% of cases; in post hoc counts, the refuser is the least tolerant member or one of the three least tolerant in 73% and 94% of the refusals, and the LCC of the same query excludes the refuser, the refused member or both in 84% and 82%, while these pairwise groups have a median of 22.5 and 5 members outside the LCC (Section A-B). Consensus is about as frequent in these groups as in LCC groups. Per run rather than per group, pairwise groups lose an interaction in 1–5% of the runs under gossip and 5–13% under concentrated weights, against 23–67% and 32–90% for (k, r)-cores at τ and size-matched SAC groups. If the tolerances that drive the dynamics deviate from those used in the search by a factor $\exp(0.3Z)$ with standard normal Z, 0–3.5% of the LCC groups lose an interaction, against 0.9–4.1% of the pairwise groups and 25–38% of the (k, r)-cores at τ.

LCC or pairwise: The LCC bound trades group size for robustness to uneven influence. Pairwise groups are at least as large by definition, since every LCC satisfies the pairwise condition: where both are proven optimal (Table VIII), the pairwise group is larger for 60%, 61%, 93% and 97% of the queries (G, B, L, O; median per-query size ratio 1.08–1.52). In exchange, LCC groups lose no accepted interaction in Table IV, whereas pairwise groups do under uneven influence, in few runs and usually for one pair at a time. This is the distinction drawn in Section III: the pairwise condition suffices when positions remain fixed (Proposition 2(a)), as in a one-off meetup, whereas for groups that interact repeatedly with unknown weights, only (iii) guarantees acceptance, as a worst case over weights rather than a prediction for typical ones.

Open groups: We add outside influence to test the closed-group assumption. When members average with accepted outside neighbours at fixed positions, Proposition 3 preserves acceptance under conditions (a) and (b). Only 4.6% and 2.6% of LCC groups on Gowalla and Brightkite meet these conditions, mainly because accepted outside friends extend K_0 beyond $\theta_{\min}$. When each accepted outside friend has weight w, against weight 1 for the member itself and for each accepted group neighbour, already at $w = 0.01$, 34% and 23% of the LCC groups lose an interaction accepted at the start, and at $w = 1$, 65% and 62%; no group that satisfies the conditions ever does. In an exploratory comparison on these two datasets, LCC groups lose an interaction less often than both size-matched SAC groups (by 3.6–16.9 pp) and pairwise groups (by 2–7 pp) at every $w > 0$, but not on Congress (Section VI-E); on the synthetic graphs, where accepted outside friends lie in the same community, no LCC group loses one even at $w = 1$. The loss is small and occurs early, within two steps, and affects a median 0.4–12% of the initially accepted interactions (Table XIX); consensus is rare for open groups of every method (at most 5.3%).

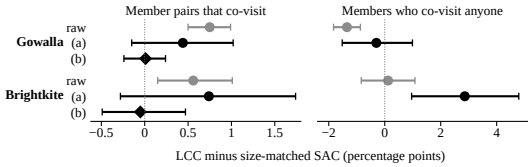


Fig. 3. LCC minus size-matched SAC in co-visit shares (pp, 95% CIs): raw, at equal size with other members (a), and size-adjusted (b, for pairs only).

D. Meetups: What Member-Specific Tolerances Change

Here θ_v denotes a travel radius. For a meetup, each member need only be able to reach every other member's home, as required by the pairwise condition in Section III. Condition (iii) is stricter, so these data measure what member-specific tolerances change and what enforcing (iii) costs, not the dynamics of Section IV. Results on the Yelp Open Dataset [56], on which we replicate this application, are omitted from this version (see the footnote on the first page). Two users *co-visit* if they check in at the same venue on the same day. Homes, tolerances and design choices use only the training and validation periods. Every test-active user in the k-core is a query, with $k = 3$: 15,829 and 5,260 queries, of which 2,469 and 462 have an LCC (median 16 and 10 members).

Tolerance violations: Groups found without tolerances leave many members with another member's home beyond their estimated travel radius (Gowalla, Brightkite; groups of at most 3,000 members): only 7% and 2% of the members of SAC groups, 29% and 42% of those of global (k, r)-cores, and 65% and 72% of those of (k, r)-cores at τ have every home within their radius, against all members of LCC and pairwise groups. Condition (iii) is stricter: on the Gowalla queries with an LCC, it holds for 39% of the pairwise groups, 57% of the (k, r)-cores at τ and 33% of the size-matched SAC groups, which are larger (median 22, 27 and 22 members, against 16).

Co-visits: Size-controlled comparisons test whether the tolerance constraint reduces co-visits relative to spatial communities of the same size; no loss is detected. Where both groups exist, LCC groups have a higher share of co-visiting member pairs than size-matched SAC on both datasets, mainly owing to size (Fig. 3 and Table XV). Where SAC groups have the LCC's size but different members (a; 610 and 108 queries), and at equal size in a regression of the difference on the log size ratio (b), no difference is measurable under a rule requiring both CIs above 0 for an advantage: both CIs contain 0 on Gowalla and Brightkite. The CIs of (a) and (b) exclude losses of more than about 0.25 and 0.5 pp. The secondary measure counts members who co-visit with anyone in the

group; at equal size it favours LCC groups on Brightkite, but not Gowalla. Against (k, r)-cores at τ, LCC groups are not measurably ahead at equal size on either dataset. Across $k \in \{2, \ldots, 5\}$ and tolerances at the 70th and 90th percentiles, the raw differences against size-matched SAC and (k, r)-cores at τ retain their sign on both datasets (as unpaired means). Within tolerance-agnostic groups, whether members who can reach everyone co-visit more depends on the dataset and the baseline once eccentricity and activity are controlled for.

Against pairwise groups: Condition (iii) may cost some co-visits relative to the pairwise condition, although size cannot be held fixed here: on Gowalla and Brightkite, the two groups are identical for 38% of the queries, and on the 60% and 61% where the pairwise group is larger, its member pairs co-visit less often (LCC minus pairwise $+1.49$ pp $[1.08, 1.95]$ and $+1.77$ pp $[1.12, 2.50]$). The size-adjusted difference (b) is nevertheless -0.85 pp $[-1.35, -0.37]$ on Gowalla and -0.29 pp $[-0.69, 0.11]$ on Brightkite; as equal-size groups are almost always identical, (b) extrapolates here (Section A-A).

An empty LCC: The absence of an LCC usually reflects a constraint already imposed by accepted friendship edges. An LCC is absent for most queries (84.4% and 91.2%), but for 95% and 97% of these, even the k-core of q's component restricted to friendships whose two ends accept each other is empty, so q has no k connected friends who accept one another; only 1.8% and 0.6% have pairwise groups but no LCC (Table XVII). An empty answer thus usually tells a user that q lacks such friends, which only a smaller k or wider tolerances could change, and tripling every tolerance gives an LCC to only 22% and 16% of these queries; conversely, shrinking every tolerance by 20% keeps an LCC for 80% of the queries that have one on both datasets, at a median 82% and 89% of its size. On Gowalla and Brightkite, an empty LCC is informative: SAC groups co-visit more when an LCC exists, by $+4.1$ pp $[3.4, 4.9]$ and $+3.1$ pp $[1.9, 4.5]$ in a post hoc regression on log size, and global (k, r)-cores show the same direction ($+0.74$ pp $[0.09, 1.42]$ and $+1.12$ pp $[-0.35, 2.51]$ without size adjustment). Without size adjustment, the share of SAC members who co-visit with anyone in the group is also higher when an LCC exists ($+0.8$ and $+5.4$ pp). On both datasets, an empty answer thus marks queries lacking a cohesive group satisfying (iii), for which groups found without tolerances also tend to meet less.

E. Positions from Votes: Two Parliaments

Opinion positions are rarely observed alongside a network. We illustrate the method using the 474 members of the 117th U.S. Congress, then repeat the main analyses on the European Parliament. Positions are the first dimension of DW-NOMINATE [57], [58], estimated from roll-call votes; θ_v is the median distance from v to the sponsors of the bills that v co-sponsored from January 2021 to February 8, 2022 [59] (median 0.145, floor 0.02, which no member reaches); and G joins members who retweeted, quoted, replied to or mentioned each other on Twitter between February 9 and June 9, 2022 [60], [61] (10,188 edges). Positions derive from votes and tolerances from co-sponsorship, rather than measured attitudes. With strong party discipline in Congress, the results concern only this setting. Tolerances do filter: 42.7% of the adjacent pairs accept each other, an LCC exists for 96.2% of the 472 queries, and its median size is 12% of that of q's k-core component.

TABLE V
TOLERANCE VIOLATIONS IN CONGRESS GROUPS ($k = 3$, 472 QUERIES).

Method	(iii) holds	Members	Pairs
LCC (Exact, Greedy, BallCore-F)	100	0	0
Pairwise	0.7	0	0
(k, r)-core at τ	15.6	4.9	1.3
Size-matched SAC	2.2	8.2	1.5
SAC	87.1	14.3	14.2
Global (k, r)-core	1.7	56.7	19.1
BallCore-R	0.9	49.6	12.9
Random, k-core, edge-accept	0	100	71.6–77.4

Premise test: The premise test asks whether co-sponsorship tolerances predict Twitter interactions beyond ideological distance. The planned hypothesis is not supported. In a logistic model of whether u interacted with v, with sender and receiver effects, a spline in the distance, and indicators of the same party, state and chamber, the coefficient of $\mathbf{1}[d(x_u, x_v) \le \theta_u]$ is -0.044 (one-sided permutation $p = 0.92$, permuting tolerances within party and chamber), and it does not improve the held-out log loss. Members with wider co-sponsorship radii interact over wider ranges (Spearman 0.71 $[0.65, 0.76]$; 0.63 $[0.56, 0.68]$ controlling for party, chamber and centrality), but the cutoff at θ_u adds nothing beyond distance and fixed effects, so this tolerance estimate is not validated on Twitter. A possible explanation, which we do not test, is the adversarial nature of some replies and mentions; Luan et al. [38] find more cross-ideological exchanges in reply ties than follow ties. Of the secondary variants, only 90th-percentile tolerances yield a significantly positive coefficient (0.32, $p = 0.001$).

Static violations and dynamics: Groups found without tolerances often contain members who face someone beyond their own tolerance (391 of the 454 size-matched SAC groups; 4.9%, 8.2% and 56.7% of the members of (k, r)-cores at τ, size-matched SAC groups and global (k, r)-cores; Table V); pairwise groups contain none by definition, but only 0.7% of them satisfy (iii). Table V also gives the groups satisfying (iii) and the member pairs with at least one refusal. In closed dynamics, no LCC group loses an interaction accepted at the start, as Theorem 1 predicts. Pairwise groups never do under equal weights or under weights proportional to the observed Twitter interactions, but 26–47% do under concentrated weights, gossip, and updates that follow one random accepted neighbour, as do 33–49% of the size-matched SAC groups and (k, r)-cores at τ; global (k, r)-cores lose an interaction in 33% of the cases even under equal weights. No LCC group satisfies the conditions of Proposition 3; in open dynamics, none loses an interaction for $w \le 0.1$, but 22%, 41% and 81% do at $w = 0.2, 0.5$ and 1, and 98% with observed weights. Unlike on

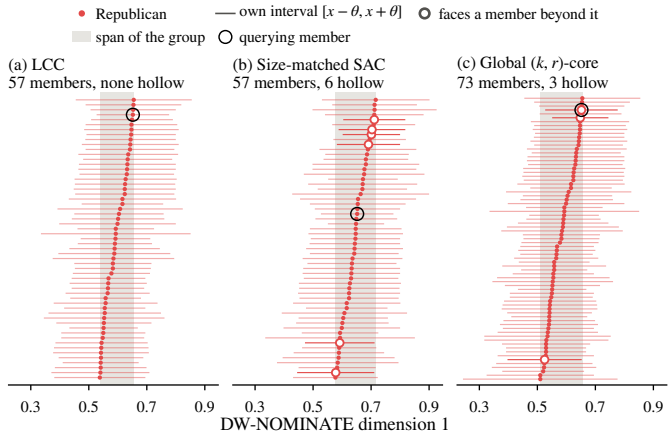


Fig. 4. Groups for one Congress query ($k = 3$), chosen by a rule fixed in advance (see text).

Gowalla and Brightkite, LCC groups here lose an interaction more often than size-matched SAC groups at every $w \geq 0.2$ (22% against 11% at $w = 0.2$), and more often than pairwise groups at $w = 0.2$ and 0.5. Because the premise test fails, these results illustrate simulated dynamics on real ideological positions; they do not establish how legislators respond to one another. Fig. 4 shows one query, chosen by a rule fixed in advance: of the 391 of 454 queries (86%) whose size-matched SAC group has a member facing another beyond its interval, the one whose LCC size is nearest the median (ties: lowest id); it illustrates Table V and adds no result.

Co-sponsorship: In the test period (June 10, 2022 to January 3, 2023), the share of member pairs who co-sponsor a bill is not measurably higher in LCC groups than in size-matched SAC groups: $+0.39$ pp $[-0.13, 0.92]$ raw, -0.30 pp $[-1.03, 0.43]$ at equal size with other members, and $+0.36$ pp $[-0.16, 0.91]$ size-adjusted; as about two thirds of the member pairs and 99.9% of the members of LCC groups co-sponsor with each other, this measure discriminates poorly.

A second parliament: On the European Parliament of 2014–16 [62] (roll-call positions, tolerances from earlier retweets), the premise test is again negative (-0.07, $p = 0.78$); only 90th-percentile tolerances give a positive coefficient (0.42, $p = 0.001$, no held-out gain), an untested hint that acceptance reaches beyond the median distance. In size-matched SAC groups, 21% of the members face someone beyond their tolerance, and 21–52% of the groups lose an initially accepted interaction under unequal influence; no LCC group does.

VII. Conclusion

Bounding group diameter by the smallest member tolerance connects community search to persistent acceptance: closed LCC groups preserve acceptance under bounded-confidence dynamics with any positive weights and avoid rejection under SJT dynamics. Natural relaxations lose these guarantees, and finding a largest LCC is NP-hard even in the plane. LatSweep searches through one threshold and returns certificates of optimality or a bound on the remaining gap. Its exact variant answers most evaluated queries faster than extracting q's k-core component, handles graphs with 1.8 billion edges, and returns an LCC of at least 0.689 times a proven upper bound on the efficiency queries stopped early.

The guarantees concern modelled closed-group dynamics. Testing observed opinion change would require longitudinal positions and tolerances that are unavailable here. In two parliaments, the U.S. Congress [58]–[60] and the European Parliament [62], tests fixed in advance found that tolerances estimated from co-sponsorship or earlier retweets do not predict who interacts on Twitter beyond ideological distance (permutation $p = 0.92$ and 0.78; Section VI-E). Outside ties void the guarantee for many real groups (Section VI-C), and once synthetic tolerances span several communities, Exact times out on many queries with an LCC (Section VI-B). For meetups, the diameter bound is conservative and may reduce co-visits relative to pairwise acceptance. Tolerances estimated from check-ins are noisy; they should not be used to label or exclude individuals. Open questions include learning tolerances, maintaining LCCs as graphs evolve, and approximating MaxLCC in fixed dimension.

Appendix A
Supplementary Results

This appendix provides the full tables for Section VI. The tables retain the runs reported in the main text; any change in sample or sweep order is stated explicitly.

A. Dynamics: All Variants and Methods

Tables VI and VII extend Table IV to all variants and methods, with the same abbreviations G, B, L and O; <1 marks shares above 0 and below 1%, and – marks runs not performed (edge-accept was not run on LiveJournal and Orkut). A group is counted in the SJT rows when it shows a new rejection. The last column of Table VI counts simulated LCC groups per variant, out of 1,000, 462, 95 and 115. Complete-graph and multi-run variants skip groups above a size cap (see Variants). Other methods have lower coverage because their larger groups are skipped more often (for gossip on Orkut, 37 of 115 (k, r)-cores at τ). For each method, the last row of both tables reports its groups under equal-weight HK on $G[H]$. The groups are those of Table IV: on Gowalla and Brightkite, 1,000 and 462 meetup queries with an LCC and 300 sampled queries without one, and on LiveJournal and Orkut, 200 default queries; the LCC, Greedy, (k, r)-cores at τ, size-matched SAC and random groups exist only for queries with an LCC, whereas the other methods may also return groups for queries without an LCC. BallCore-F returns subsets satisfying (iii), and no BallCore-F group loses an interaction in a closed variant. We omit the k-core component because almost all its groups exceed 5,000 members, the largest group size we simulate.

Variants: HK (1) runs with equal weights, also for 3,000 steps with a stopping rule 10^6 times stricter, or with weights drawn in each run from a Dirichlet distribution with $\alpha = 1$ or $\alpha = 0.1$. In *follow one neighbour*, each member moves

TABLE VI

GROUPS (%) THAT LOSE AN INTERACTION ACCEPTED AT THE START IN ALL 21 VARIANTS: METHODS THAT USE TOLERANCES.

	LCC (Exact)				Greedy				BallCore-R				Pairwise				Edge-accept				LCC groups
Variant	G	B	L	O	G	B	L	O	G	B	L	O	G	B	L	O	G	B	L	O	G/B/L/O
Closed group, interactions on $G[H]$																					
HK, equal weights	0	0	0	0	0	0	0	0	<1	0	0	0	0	0	0	0	67	63	–	–	1,000/462/95/115
HK, equal, 3,000 steps	0	0	0	0	0	0	0	0	<1	0	0	0	0	0	0	0	67	63	–	–	1,000/462/95/115
HK, random weights, $\alpha = 1$	0	0	0	0	0	0	0	0	19	7	0	<1	1	<1	0	0	78	77	–	–	1,000/462/95/115
HK, random weights, $\alpha = 0.1$	0	0	0	0	0	0	0	0	62	56	64	76	40	33	32	40	85	85	–	–	1,000/462/95/115
Follow one neighbour	0	0	0	0	0	0	0	0	45	31	3	12	7	8	0	<1	81	80	–	–	1,000/462/95/115
Gossip (DW)	0	0	0	0	0	0	0	0	54	38	28	58	17	11	3	7	82	82	–	–	1,000/462/95/115
SJT, new rejection	0	0	0	0	0	0	0	0	29	15	49	82	<1	0	0	<1	74	73	–	–	1,000/462/95/115
Closed group, complete interaction graph																					
HK, equal weights	0	0	0	0	0	0	0	0	<1	0	0	0	0	0	0	0	57	62	–	–	1,000/462/95/115
HK, random weights, $\alpha = 1$	0	0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	57	73	–	–	931/462/95/115
HK, random weights, $\alpha = 0.1$	0	0	0	0	0	0	0	0	29	39	11	12	13	23	5	4	76	85	–	–	931/462/95/115
Follow one neighbour	0	0	0	0	0	0	0	0	51	46	10	33	30	17	1	<1	72	82	–	–	931/462/95/115
Gossip (DW)	0	0	0	0	0	0	0	0	47	44	7	43	27	21	6	17	66	72	–	–	907/462/63/71
SJT, new rejection	0	0	0	0	0	0	0	0	16	17	65	87	<1	0	0	2	70	72	–	–	1,000/462/95/115
Assumptions violated																					
Open group, $G[H]$	65	62	0	0	63	63	0	0	74	70	0	0	72	67	0	0	83	85	–	–	1,000/462/95/115
Open group, complete	61	57	0	0	57	57	0	0	67	62	0	0	65	60	0	0	78	86	–	–	1,000/462/95/115
Stubborn outsiders, $\lambda = 0.01$	91	94	1	<1	90	94	1	1	93	97	3	2	93	97	0	2	97	100	–	–	1,000/462/95/115
Stubborn outsiders, $\lambda = 0.05$	98	99	49	33	98	99	39	26	98	100	73	46	98	100	65	38	99	100	–	–	1,000/462/95/115
Stubborn outsiders, $\lambda = 0.1$	99	100	91	71	98	100	88	68	99	100	93	84	99	100	93	80	99	100	–	–	1,000/462/95/115
Tolerance noise, $\nu = 0.1$	0	0	0	0	0	0	0	0	3	1	0	0	0	0	0	0	70	69	–	–	1,000/462/95/115
Tolerance noise, $\nu = 0.2$	0	<1	0	0	<1	0	0	0	16	5	<1	3	0	<1	0	0	73	73	–	–	1,000/462/95/115
Tolerance noise, $\nu = 0.3$	2	3	0	3	2	4	0	1	31	21	6	12	3	4	<1	2	76	75	–	–	1,000/462/95/115
Groups (HK, equal, $G[H]$)	1,000	462	95	115	887	447	83	93	1,004	467	107	130	1,003	464	107	129	1,011	471	–	–	

TABLE VII

AS TABLE VI: METHODS THAT IGNORE TOLERANCES.

	(k,r)-core at τ				Global (k,r)-core				SAC				Size-matched SAC				Random			
Variant	G	B	L	O	G	B	L	O	G	B	L	O	G	B	L	O	G	B	L	O
Closed group, interactions on $G[H]$																				
HK, equal weights	15	17	4	13	60	33	7	6	18	32	1	3	5	8	1	2	37	27	0	0
HK, equal, 3,000 steps	15	17	4	13	60	33	7	6	18	32	1	3	5	8	1	2	37	27	0	0
HK, random weights, $\alpha = 1$	26	35	43	68	77	81	53	77	26	43	22	42	21	29	38	59	55	45	21	26
HK, random weights, $\alpha = 0.1$	41	44	91	97	86	89	99	99	41	56	78	84	52	56	89	98	64	53	75	69
Follow one neighbour	33	39	58	78	82	81	73	86	30	45	26	41	35	35	48	67	55	45	15	24
Gossip (DW)	34	40	69	85	83	82	81	93	32	47	38	55	39	40	63	86	57	47	39	31
SJT, new rejection	25	31	92	98	75	71	99	99	28	47	61	83	21	31	85	97	90	89	98	97
Closed group, complete interaction graph																				
HK, equal weights	18	23	33	54	59	50	39	56	15	31	13	25	8	14	15	34	53	40	2	2
HK, random weights, $\alpha = 1$	18	31	37	69	62	68	43	76	18	40	17	42	17	28	29	65	65	61	29	37
HK, random weights, $\alpha = 0.1$	29	45	89	95	79	87	97	99	31	51	74	83	37	47	88	98	74	68	81	84
Follow one neighbour	36	43	70	89	77	86	78	96	30	44	40	58	45	48	63	88	67	61	39	39
Gossip (DW)	27	42	54	78	73	81	74	84	28	43	34	53	41	46	50	80	66	61	33	30
SJT, new rejection	24	29	92	100	71	75	99	99	23	44	67	85	19	30	93	97	78	78	93	97
Assumptions violated																				
Open group, $G[H]$	75	80	4	12	84	84	5	9	61	74	4	11	73	79	5	14	49	37	0	<1
Open group, complete	73	81	33	55	80	87	39	57	58	72	14	31	70	80	15	44	60	46	2	3
Stubborn outsiders, $\lambda = 0.01$	96	99	52	68	97	99	64	76	92	97	24	29	95	99	38	45	80	77	12	9
Stubborn outsiders, $\lambda = 0.05$	98	100	83	90	99	100	97	97	97	99	59	60	98	100	79	82	89	86	48	35
Stubborn outsiders, $\lambda = 0.1$	99	100	96	97	99	100	99	100	98	99	80	84	99	100	95	97	91	88	80	70
Tolerance noise, $\nu = 0.1$	17	22	16	32	65	58	15	20	17	33	4	10	9	13	6	10	44	32	0	<1
Tolerance noise, $\nu = 0.2$	23	27	21	39	70	65	24	34	20	37	8	12	15	19	9	14	48	37	0	3
Tolerance noise, $\nu = 0.3$	28	35	25	38	74	73	31	40	26	39	10	16	23	29	15	23	51	42	1	2
Groups (HK, equal, $G[H]$)	1,000	462	95	115	994	428	150	162	1,243	745	153	178	1,000	462	95	115	1,000	462	95	115

halfway towards one accepted neighbour drawn at random in each step; in gossip [32], a random edge interacts, and each end that accepts the other moves 0.3 of their distance towards it, for 100 rounds of $|E(H)|$ such steps (50 rounds on complete graphs). SJT (2) uses equal weights, $\theta_v^{\mathrm{acc}} = 0.5\,\theta_v$, $\theta_v^{\mathrm{rej}} = \theta_v$ and $\beta = 0.05$; there we report new rejections, the event that Theorem 2 excludes. Randomized variants run 10 times (gossip on complete graphs 5 times), and a group counts if any run loses an interaction. On complete interaction graphs, every member hears every other; these variants run on groups of at most 1,000 members (equal weights, SJT, open), 300 (random weights, follow) or 100 (gossip), and the randomized variants on $G[H]$ on groups of at most 2,000 members. The last three blocks test sensitivity to departures from the assumptions in Section IV. In open groups, members also average with the outside neighbours they accept, which stay fixed, with the same weight as accepted members ($w = 1$ in Section VI-C; at most 200 outside neighbours per member, sampled at random); stubborn outsiders pull each member with weight λ towards the fixed mean of all its outside neighbours, near or far; with tolerance noise, the dynamics use $\theta_v \exp(\nu Z_v)$ with standard normal Z_v, whereas the groups were found with θ_v.

Observations: In closed variants, no group returned by a method guaranteeing (iii) (LCC, Greedy) loses an interaction; a tenfold longer run leaves every entry unchanged. BallCore-R, whose sets need not satisfy (iii), loses interactions under uneven influence in up to 76% of its groups on $G[H]$. On complete interaction graphs, pairwise groups also lose interactions under uneven influence (up to 30%), and the (k, r)-cores at τ already under equal weights (18–54%). With stubborn outsiders, at least 77% of the groups of every method lose an interaction on the meetup graphs, whose outside friends lie in all directions; on the synthetic graphs at $\lambda = 0.01$, about 1% of the LCC groups do, against 52–68% of the (k, r)-cores at τ. Up to $\nu = 0.2$, tolerance noise barely affects LCC and pairwise groups (below 1%), while 16–39% of (k, r)-cores at τ lose an interaction.

LCC or pairwise: Table VIII provides the results for the paragraph "LCC or pairwise" of Section VI-C. It counts queries for which both LCC and pairwise groups exist and are proven optimal, the share where the pairwise group is strictly larger, and the median pairwise-to-LCC size ratio across these queries. The other columns give the pairwise groups (%) that lose an interaction accepted at the start, for the groups of Table IV: in any of 10 runs, as in Table IV, and on average per run; no LCC group ever does. Pairwise groups are strictly larger than LCCs for most queries, and under uneven influence they lose an accepted interaction in a few of their runs: on average 5–13% of the runs under concentrated weights, 1–5% under gossip, and at most 2% when each member follows one neighbour. On Gowalla and Brightkite, the pairwise group is the LCC itself for 938 of the 978 and 177 of the 181 queries on which the two have the same size (the paired co-visit difference on these is $+0.14$ pp $[-0.03, 0.36]$ and 0.00 pp), and on the 1,491 and 281 queries on which it is larger (median 52 against 34 and 17 against 12 members), the LCC's pair rate

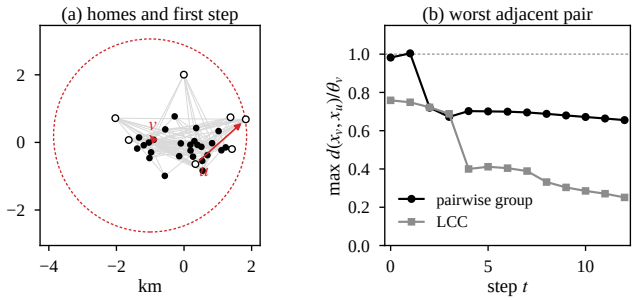


Fig. 5. A refusal in a Gowalla pairwise group: (a) homes and first moves; (b) worst adjacent pair per step.

is higher by $+1.49$ pp $[1.08, 1.95]$ and $+1.77$ pp $[1.12, 2.50]$. The size-adjusted difference (b) of Section VI-D, a linear regression of the difference on the log size ratio, gives intercepts of -0.85 pp $[-1.35, -0.37]$ and -0.29 pp $[-0.69, 0.11]$. The relation is far from linear: on Gowalla, the mean difference is $+0.14$ pp for log size ratios in $[0, 0.25)$, $+1.2$ and $+0.8$ pp in $[0.25, 0.5)$ and $[0.5, 1)$, and $+7.6$ and $+27$ pp for the 67 and 32 queries with log ratios in $[1, 2)$ and above 2, which set the slope; without the latter 99 queries, the intercept is $+0.10$ pp. On Brightkite, the intercept stays negative without the queries of log ratio at least 1 (-0.38 pp). The fitted line also misses the equal-size queries for size-matched SAC groups ($+0.01$ against $+0.19$ pp on Gowalla, -0.05 against $+0.30$ pp on Brightkite), so (b) is a coarse adjustment for every baseline, and with pairwise groups, whose equal-size groups are almost all the LCC itself, it rests on extrapolation. Results on Yelp are omitted in this version (see the footnote on the first page).

B. Refusals in Pairwise Groups

Table IX examines the refusals behind the column of concentrated weights in Table IV on Gowalla and Brightkite. For the queries with an LCC, we replayed every run of the pairwise and LCC groups of that column with the same random streams and an instrumented copy of the simulator that records, for each directed interaction (v accepts u) accepted at the start, the first step at which it is refused; the replay reproduces the stored results of every group and run exactly, and no LCC group ever refuses an interaction. The table counts refused interactions over all runs; a refuser counts as the least tolerant member if its tolerance equals the group's minimum (ties included), and as one of the three least tolerant if it is the least tolerant or its average rank from the least tolerant is at most 3 (tied members share their average rank). The counts for the three least tolerant members and for either participant lying outside the LCC were added after the analysis plan was fixed.

At the start, refused interactions are not close to tolerance (median d/θ_v of 0.55, and at least 0.9 for only 9% and 12%). Refusal occurs mostly at the first step. During the run, the largest distance excess over θ_v has a median of 8% and 10%; by the end, all but about 1% are accepted again. The refuser is one of the three least tolerant members of the group in 73% and 94% of the refused interactions and is (or ties for)

TABLE VIII
LCC AGAINST PAIRWISE GROUPS: SIZES, AND PAIRWISE GROUPS (%) THAT LOSE AN ACCEPTED INTERACTION ON $G[H]$.

		Pairwise vs. LCC		Concentrated		Gossip		Follow one	
Graph	Queries	Larger (%)	Size ratio	Any	Per run	Any	Per run	Any	Per run
Gowalla	2,469	60.4	1.08	40.4	13.2	16.6	5.1	6.9	1.9
Brightkite	462	60.8	1.09	33.4	8.1	10.6	2.9	7.8	1.5
LiveJournal	87	93.1	1.33	31.8	4.8	2.8	0.8	0.0	0.0
Orkut	102	97.1	1.52	39.5	7.8	7.0	1.7	0.8	0.2

TABLE IX
REFUSED INTERACTIONS IN PAIRWISE GROUPS UNDER CONCENTRATED WEIGHTS ($d = d(x_v, x_u)$; v REFUSES u).

	Gowalla	Brightkite
Pairwise groups with a refusal	402 of 1,000	153 of 462
Refused interactions, all runs	2,200	592
Initial d/θ_v, median [quartiles]	0.55 [0.36, 0.73]	0.55 [0.28, 0.77]
Initial $d/\theta_v \geq 0.9$ (%)	8.9	11.8
Step of refusal, median	1	1
Largest d/θ_v, median	1.08	1.10
Still refused at the end (%)	1.2	1.0
Refuser is the least tolerant member (%)	45.3	73.3
Refuser among the three least tolerant (%)	72.9	93.9
Refuser outside the LCC (%)	69.1	71.1
Refuser or refused outside the LCC (%)	83.7	82.4
Pairwise members outside the LCC, median	22.5	5

the least tolerant one in 45% and 73%; the least tolerant member is the one that bounds the diameter in (iii). In 84% and 82% of the refused interactions, the LCC of the same query excludes the refuser, the refused member or both, while the pairwise groups with a refusal have a median of 22.5 and 5 members outside the LCC. These refusals use simulated influence weights. Closed Congress dynamics with weights proportional to observed interactions produce no loss in any pairwise group (Section VI-E).

Fig. 5 shows the case chosen by a rule fixed in advance: among the 250 Gowalla groups of at most 80 members with a refusal, the group whose size is nearest the median of 33 (ties: lowest query), its first run with a refusal, and its earliest refused interaction. Positions are shown relative to the group centroid. Member u ($\theta_u = 15.3$ km, outside the LCC) puts 99.8% of its weight on one neighbour, also outside the LCC, and moves 2.0 km in the first step; its distance to v ($\theta_v = 2.86$ km, the second least tolerant of the 33 members, in the LCC of 27 members) grows from 0.50 to $1.004\,\theta_v$ and is back within θ_v from the second step on. For the same query and run index, every adjacent pair in the LCC remains within 0.76 times its smaller tolerance.

C. Heterogeneity of Tolerances

Table X provides further results for the paragraph on heterogeneity in Section VI-B. It uses 200 default queries per setting and the count-order sweep (Exact 60 s, the other searches 10 s). For each σ, it gives the queries (%) with an LCC and those reaching the 60 s limit, out of all; for the other methods, columns (iii) holds, Members and Size/LCC give the groups (%) satisfying (iii), the mean share (%) of members who face another member beyond their own tolerance, and the median size relative to the LCC where both exist; Global (k, r) is the greedy (k, r)-core search with the median tolerance. With equal tolerances, every (k, r)-core at τ and every pairwise group satisfies (iii), and their median size equals the LCC's; Exact reaches the limit on 45% of the Orkut queries there, the setting in which the LCC is a maximum (k, r)-core. As σ grows, fewer queries have an LCC, and the (k, r)-cores at τ grow to 1.25–4.21 times the LCC's size (median) while at most 9% of them satisfy (iii). The reported time-limit shares use the count-order sweep; the best-first sweep is used in Table II.

D. Synthetic Robustness

Table XI provides further results for the paragraph on the tolerance scale and the proven share in Section VI-B, with 200 queries per setting and a 60 s limit. Column LCC counts the queries with an LCC; the next columns give shares (%) of these queries: proven, reaching the limit, and stopped at the cap on searched sets; the last column gives, for the queries reaching the limit, the median ratio (%) of the returned LCC to the proven upper bound. Here the mean tolerance is c times the median distance between friends (106.9 on LiveJournal and 126 on Orkut, mostly between communities), without calibration on Exact; $c = 0.5$ already corresponds to scales s of 9.4 and 11.1, against the calibrated 1.5 and 1.25. We also draw queries uniformly from the whole k-core instead of from groups of at least 20 members, and move a share μ of the vertices to a random community. Where tolerances reach the scale between communities ($c \geq 0.75$ on Orkut, $c \geq 1$ on LiveJournal), 35–99% of the queries with an LCC reach the time limit and the bounds for these queries are loose; in the other settings, Exact proves every query.

E. Caps and Time Limit at the Scale between Communities

Table XII varies one parameter of Section V-C at a time on the three settings of Table XI where Exact starts to time out, with the same instances and queries: the cap on searched sets (default 3,000), the cap on the root bound (default 20,000), the number of thresholds Greedy tries for the initial incumbent (default 20), and, for the queries left unproven at the default (C0), a time limit of 300 s instead of 60 s. Raising a cap changes the proven share by at most +1.1 pp (two queries on Orkut at $c = 0.75$); with a larger cap on searched sets, the queries that stopped at that cap under C0 (0.6–1.1%) reach the time limit instead. Letting Greedy try all thresholds lowers the share by 3.3 and 2.3 pp on LiveJournal at $c = 1$ and

TABLE X
HETEROGENEITY SWEEP ON LIVEJOURNAL (L) AND ORKUT (O), COUNT-ORDER SWEEP.

	With LCC		Exact limit		(k,r)-core at τ						Pairwise		Global (k,r)	
					(iii) holds		Members		Size/LCC		(iii) holds		Members	
σ	L	O	L	O	L	O	L	O	L	O	L	O	L	O
0.0	76.5	85.5	0.5	45.0	100.0	100.0	0	0	1.00	1.00	100.0	100.0	0	0
0.2	70.0	74.5	14.0	26.5	8.6	4.7	12	23	1.25	1.50	4.1	0.6	27	38
0.4 (default)	47.5	57.5	4.0	7.0	4.2	0.0	31	43	1.80	2.19	5.6	2.3	40	44
0.6	39.0	44.0	0.5	3.0	2.6	1.1	43	52	2.54	2.93	7.1	3.1	45	47
0.8	30.5	33.0	0.0	0.0	1.6	0.0	49	61	2.72	3.87	9.0	2.7	47	49
1.0	21.0	27.5	0.0	0.0	2.4	1.8	55	65	3.50	4.21	11.1	6.9	49	48

TABLE XI
SYNTHETIC SETTINGS: SCALE, QUERY SAMPLING AND COMMUNITY MIXING.

Graph	Setting	LCC	Proven (%)	Limit (%)	Size cap (%)	Solution/ bound (%)
LiveJournal	$c = 0.5$	153	100.0	0.0	0.0	–
LiveJournal	$c = 0.75$	153	100.0	0.0	0.0	–
LiveJournal	$c = 1$	153	64.1	35.3	0.7	0.33
LiveJournal	$c = 1.5$	150	15.3	84.0	0.7	0.04
LiveJournal	$c = 2$	119	3.4	95.0	1.7	0.03
Orkut	$c = 0.5$	177	100.0	0.0	0.0	–
Orkut	$c = 0.75$	177	50.8	48.0	1.1	0.91
Orkut	$c = 1$	177	19.8	79.7	0.6	0.18
Orkut	$c = 1.5$	161	3.1	96.3	0.6	0.40
Orkut	$c = 2$	177	0.6	99.4	0.0	7.6
LiveJournal	q from the k-core	91	100.0	0.0	0.0	–
Orkut	q from the k-core	92	100.0	0.0	0.0	–
LiveJournal	mixing $\mu = 0.1$	80	100.0	0.0	0.0	–
LiveJournal	mixing $\mu = 0.2$	68	100.0	0.0	0.0	–
LiveJournal	mixing $\mu = 0.4$	42	100.0	0.0	0.0	–
Orkut	mixing $\mu = 0.1$	103	100.0	0.0	0.0	–
Orkut	mixing $\mu = 0.2$	83	100.0	0.0	0.0	–
Orkut	mixing $\mu = 0.4$	51	100.0	0.0	0.0	–

TABLE XII
PROVEN SHARE (%) OF THE QUERIES WITH AN LCC WHEN ONE CAP OR THE TIME LIMIT CHANGES (200 QUERIES PER SETTING).

		Proven (%) with				300 s: proven /
Setting (LCC)	C0	cap 20,000	cap 50,000	root 100,000	Greedy all τ	unproven under C0
LiveJournal $c = 1$ (153)	64.1	64.1	64.1	64.1	60.8	5 / 55
Orkut $c = 0.75$ (177)	50.8	51.4	52.0	50.8	48.6	1 / 87
Orkut $c = 1$ (177)	19.8	20.3	20.3	19.8	19.8	0 / 142

Orkut at $c = 0.75$, because the longer initial phase consumes part of the limit. Under a 300 s limit, Exact proves 5 of 55, 1 of 87 and 0 of 142 queries. Across configurations, the median returned-LCC-to-bound ratio remains 0.18–1.3%. Every configuration returns the same optimum wherever it proves one, in agreement with Table XI; the queries with an LCC have a median of 1,218 (LiveJournal), 8,360 and 210,146 (Orkut, $c = 0.75$ and 1) candidate vertices, and each engine process uses at most 2.7 GB.

F. Sweep against a Single Threshold

Table XIII runs our engine on one threshold only, as a maximum (k,r)-core search containing q without personal tolerances, at the median tolerance of all vertices (as the global baseline) and at the threshold τ^* of the optimal LCC, which the single-threshold search does not know in advance; the latter is run only on queries with an LCC. Time ratios are computed per query with an LCC, flooring both times at 1 ms; the last column reports the share of these queries for which the sweep is faster. Query times exclude index construction, which the single-threshold search does not use. The index is built once per process; in this run it took 5.7 s on LiveJournal and 14.3 s on Orkut (a slower run than the 9.7 s of Section VI-B). On LiveJournal, where the sweep answers most queries in under 1 ms, amortizing this cost over 200 queries would raise the median ratio to about 39. The non-empty single-threshold answers satisfy (iii) on 0% of the queries at the median tolerance and on 4.2% (LiveJournal) and 0% (Orkut) at τ^*.

G. Unproven Efficiency Queries

Table XIV lists the unproven queries of Section VI-B, with the size of the returned LCC, the proven upper bound UB, their ratio (truncated), the candidate vertices (those with a non-empty survival interval), the thresholds evaluated out of all,

TABLE XIII
THE SWEEP AND SINGLE-THRESHOLD (k, r)-CORE SEARCHES ON THE 200 DEFAULT QUERIES OF TABLE III (60 S LIMIT).

		Queries (of 200)			Median	Time ratio, sweep / single	
Graph	Search	Proven non-empty	Proven empty	Limit	time (ms)	Median	Sweep faster (%)
LiveJournal	Sweep (Exact)	95	105	0	0.13	–	–
LiveJournal	(k, r)-core at median θ	131	50	19	2.1	0.89	55
LiveJournal	(k, r)-core at τ^* (95 queries)	79	–	16	6.2	0.86	56
Orkut	Sweep (Exact)	115	85	0	3.2	–	–
Orkut	(k, r)-core at median θ	96	33	71	959	0.065	92
Orkut	(k, r)-core at τ^* (115 queries)	78	–	37	2,586	0.135	94

TABLE XIV
THE NINE EFFICIENCY QUERIES THAT EXACT LEAVES UNPROVEN WITHIN 60 S.

Graph	Setting	LCC	UB	LCC/UB	Candidates	Thresholds	Nodes
Friendster	default	147	212	0.693	587	22/25	519,058
Friendster	default	305	336	0.907	560	83/84	263,320
Friendster	default	374	398	0.939	738	69/78	102,156
LiveJournal	2× scale	552	578	0.955	803	34/35	60,421
Orkut	2× scale	719	732	0.982	918	154/851	70,370
Pokec	default	169	245	0.689	622	77/83	539,455
Pokec	default	179	246	0.727	645	73/79	417,817
Pokec	default	189	253	0.747	578	323/484	551,931
Pokec	default	369	394	0.936	787	379/539	199,822

TABLE XV
THE VALUES OF FIG. 3 (PP, 95% BOOTSTRAP CIS).

	Gowalla	Brightkite
Pairs, raw	+0.75 [0.50,0.99]	+0.56 [0.15,1.01]
(a) same size	+0.44 [−0.15,1.02]	+0.74 [−0.28,1.74]
(b) adjusted	+0.01 [−0.24,0.24]	−0.05 [−0.49,0.47]
Members, raw	−1.35 [−1.83,−0.87]	+0.12 [−0.84,1.08]
(a) same size	−0.30 [−1.52,0.99]	+2.86 [0.96,4.79]
Queries, all / (a)	2,469 / 610	462 / 108

and the branch-and-bound nodes. Every candidate set $C(\tau)$ is a subset of the candidate vertices, of which these queries have at most 918 and no efficiency query has more than 1,502, so the size caps of Section V-C never apply; the search explores 60,421–551,931 nodes without settling all thresholds. The bound pass after the limit, which query times exclude, takes at most 4 µs on these queries; on all other efficiency queries, it does not run.

H. Meetups

Table XV gives the values plotted in Fig. 3: LCC minus size-matched SAC where both exist, in the share of member pairs that co-visit and of members who co-visit with anyone in the group: raw, on queries where SAC returns a group of the LCC's size with other members (a), and as the intercept of a regression on the log size ratio (b). The Yelp column is omitted in this version (see the footnote on the first page).

I. Yelp on a Second Time Split

This Yelp robustness check is omitted in this version, with Table XVI (see the footnote on the first page).

TABLE XVI
YELP, LCC MINUS SIZE-MATCHED SAC (PP, 95% BOOTSTRAP CIS) ON THE TWO TIME SPLITS.

Omitted in this version; see the footnote on the first page.

J. Queries without an LCC and Tolerance Scaling

Table XVII provides the classification for the paragraph on empty LCCs in Section VI-D. At the median tolerance and on all queries, a query without an LCC is in class A if the k-core of q's component restricted to friendships whose ends accept each other is empty, in B if it is not but no pairwise group exists, in C if a pairwise group exists but no LCC, and in U if the LCC or pairwise search is not proven; columns A–U give shares (%) of the queries without an LCC (Congress: counts). The next two columns give the share of these queries that gain an LCC when every tolerance is the 70th or 90th percentile of the user's distances, and the last five the share of queries without an LCC at the 90th percentile and its classes; raised tolerances use the 2,000 sampled queries of the parameter study, and on Congress the two columns count all 18 queries without an LCC. Raising tolerances to the 90th percentile gives an LCC to 605 of 1,698 and 606 of 1,822 sampled queries without one. Queries of classes B and C almost all gain an LCC at the 90th percentile (B: 45 of 45 and 50 of 51; C: 41 of 41 and 8 of 8 on Gowalla and Brightkite), but only about a third of class A do (519 of 1,612 and 548 of 1,763). Newly obtained LCCs have median sizes of 21.5 and 13 members at the 70th percentile, and 103 and 42 at the 90th. The Yelp row is omitted in this version (see the footnote on the first page).

Table XVIII provides the full tolerance-scaling results for the same paragraph, on the same queries at every s with Exact limited to 60 s. It gives the queries (%) with an LCC, the queries (%) proven within the limit, the median LCC size, and the queries without an LCC by class (Table XVII). For $s < 1$, column Kept gives the share (%) of the queries with an LCC at $s = 1$ that keep one, with the median size ratio of the kept LCCs in parentheses; for $s > 1$, column Gained gives the share of the queries without an LCC at $s = 1$ that gain one. The last two columns report the mean co-visiting share of member pairs in LCC groups and its difference from $s = 1$, for queries with an LCC at both scales (pp, 95% bootstrap CI). Whenever both solutions are proven, increasing s never shrinks an LCC; the Yelp block is omitted (see the footnote on the first page).

TABLE XVII
QUERIES WITHOUT AN LCC ($k = 3$) BY CAUSE, AND WITH RAISED TOLERANCES.

			Without LCC, median tolerance (%)				Solved when raised (%)		90th percentile, sample (%)				
Data	Queries	No LCC	A	B	C	U	70th	90th	No LCC	A	B	C	U
Gowalla	15,829	13,360 (84.4%)	95.0	3.2	1.8	0	15.0	35.6	54.7	85.1	11.8	3.1	0
Brightkite	5,260	4,798 (91.2%)	96.7	2.6	0.6	0	10.0	33.3	60.8	81.7	14.3	3.9	0
Congress	472	18 (3.8%)	14	2	2	0	12/18	16/18	–	–	–	–	–

TABLE XVIII
TOLERANCE SCALING: EVERY TOLERANCE MULTIPLIED BY s.

				Queries without LCC						Pairs that co-visit	
s	LCC (%)	Proven (%)	Size	A	B	C	U	Kept (%)	Gained (%)	Share (%)	vs. $s = 1$ (pp)
Gowalla (15,829 queries)											
0.5	7.9	100.0	9	14,204	244	129	0	50.7 (0.58)	–	11.12	+3.73 [3.15,4.40]
0.67	10.5	100.0	10	13,617	392	159	0	67.3 (0.67)	–	9.44	+1.86 [1.52,2.26]
0.8	12.5	100.0	11	13,237	424	194	0	80.0 (0.82)	–	8.84	+1.30 [1.03,1.59]
1	15.6	100.0	16	12,691	423	246	0	–	–	8.42	–
1.25	19.3	100.0	22	11,942	557	270	0	–	4.4	7.33	−1.29 [−1.57,−1.03]
1.5	22.7	100.0	27	11,431	552	252	0	–	8.4	6.67	−2.18 [−2.51,−1.87]
2	27.3	100.0	40	10,657	583	274	0	–	13.8	5.93	−3.09 [−3.50,−2.69]
3	33.9	100.0	53	9,683	563	224	0	–	21.6	5.13	−3.84 [−4.28,−3.41]
Brightkite (5,260 queries)											
0.5	4.5	100.0	8	4,970	48	6	0	51.1 (0.78)	–	8.58	+1.18 [0.34,2.13]
0.67	5.9	100.0	8	4,849	77	22	0	67.5 (0.80)	–	7.56	+0.68 [0.18,1.21]
0.8	7.1	100.0	8	4,762	106	21	0	80.3 (0.89)	–	6.44	+0.24 [−0.08,0.59]
1	8.8	100.0	10	4,640	127	31	0	–	–	5.67	–
1.25	11.0	100.0	12	4,473	146	62	0	–	2.4	4.32	−1.35 [−1.91,−0.85]
1.5	12.9	100.0	13	4,326	172	85	0	–	4.5	3.60	−2.11 [−2.74,−1.52]
2	17.1	100.0	16	4,088	175	95	0	–	9.2	3.15	−3.19 [−4.18,−2.32]
3	23.5	100.0	24	3,746	186	92	0	–	16.1	2.52	−3.75 [−4.80,−2.79]

TABLE XIX
OPEN LCC GROUPS: LOSSES AND FINAL OPINION CLUSTERS.

	Gowalla			Brightkite			Congress			
w	0.01	0.1	1	0.01	0.1	1	0.01	0.1	1	obs.
Groups that lose an interaction (%)	34.1	48.4	65.2	22.5	37.7	62.1	0	0	80.6	98.2
lost share (median, %)	4.0	8.5	11.0	0.4	11.8	10.8	–	–	2.1	5.3
first loss (median step)	1	1	1	1	1	1	–	–	2	1
Largest final cluster (mean, %)	23.5	17.7	16.8	24.0	18.0	18.8	71.6	11.8	5.2	3.9
Largest cluster, LCC − size-matched SAC (pp)	−1.0	+0.8	+0.9	−1.4	+0.4	+0.3	−4.7	−1.5	−0.3	−0.4
95% CI	[−1.7,−0.3]	[0.4,1.2]	[0.5,1.2]	[−2.4,−0.3]	[−0.0,0.9]	[−0.1,0.7]	[−6.4,−3.1]	[−2.3,−0.8]	[−0.4,−0.1]	[−0.5,−0.3]

Larger tolerances give more and larger LCCs whose pairs co-visit less, and smaller ones fewer and smaller LCCs whose pairs co-visit more; all paired differences exclude 0 except on Brightkite at $s = 0.8$. Class A dominates queries without an LCC at every s. Shrinking tolerances also protects against noisy estimates: with tolerance noise $\nu = 0.3$ in equal-weight HK on $G[H]$ (10 runs; a separate run on all LCC groups, with different random draws from those in Table VI), 1.17% $[0.82, 1.68]$ of the Gowalla and 3.68% $[2.31, 5.81]$ of the Brightkite LCC groups at $s = 1$ lose an interaction accepted at the start, against 0% $[0, 0.19]$ and 1.08% $[0.42, 2.74]$ at $s = 0.8$ (Wilson intervals; without noise, none does).

K. Open Groups at Group Level

Table XIX covers the LCC groups of Table VI on Gowalla and Brightkite and the 454 LCC groups on Congress, when each accepted outside friend has weight w or, on Congress, a weight proportional to the observed interactions (obs.). It gives the lost share of the interactions accepted at the start, in groups that lose one, and the largest final opinion cluster, by single linkage at 10^{-3} of the initial diameter after at most 300 steps (closed groups: 99.7–100%); its last rows give LCC minus size-matched SAC in this cluster, as an exploratory comparison (closed groups: +1.6, +1.7 and 0 pp). Interaction loss and group-wide convergence are separate outcomes: open groups lose only a small share of interactions, usually early. Except on Congress at $w = 0.01$, open groups also fail to reach a common opinion whether or not they lose an interaction: members stay spread between the outside friends they average with, so the largest final cluster holds a minority of the members of LCC groups that lose no interaction as well (22–35% on Gowalla and Brightkite), and on Congress at $w = 0.1$, where no LCC group loses an interaction, only 11.8%. Pairwise groups, size-matched SAC groups and (k, r)-cores at τ also have largest final clusters of 15–28%. Paired

differences relative to size-matched SAC vary in sign, so open LCC groups have no consistent group-level advantage.

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